MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, INTERNALLY TRIMMED ANALOG MULTIPLIER, HYBRID AND MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic and hybrid silicon, internally trimmed analog multipliers. Two product assurance classes and a choice of case outlines and lead finishes are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Maximum multiplying error
01	4 quadrant analog multiplier	1%
02	4 quadrant analog multiplier	2%
03	4 quadrant analog multiplier	4%
04	4 quadrant analog multiplier	4%

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline, MIL-M-38510, appendix C
I	A-2 (10-lead can)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package

1.3 Absolute maximum ratings.

Supply voltage	±22 V
Internal power dissipation	500 mW
Input voltage 1/	
X, Y, V_{OS}, and Z	$\pm V_S$
Storage temperature	-65°C to +150°C
Output short circuit to ground	Indefinite
Lead temperature (soldering, 60 seconds)	300°C
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage range	13.5 V dc to ±18 V dc
X input voltage DIFF/CM	±10 V dc maximum
Y input voltage DIFF/CM	±10 V dc maximum
Z pin:	
Device types 01, 02, and 04	±10 V dc
Device type 03, (Z pin connected to output pin)	±10 V dc
Ambient operating temperature range	-55°C to +125°C

1/ Maximum input voltage is zero when supplies are turned off.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Docuemnt Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
Can	I	333 mW @ T_A = +125°C	25°C/W	+150°C/W
Dual-in-line	C	400 mW @ T_A = +125°C	22°C/W	+85°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, citred in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. For device type 04 only, polymeric adhesive materials are permitted in accordance with MIL-M-38510, appendix G.

3.2.1 Case outline. The case outline shall be as specified in 1.2.3.

3.2.2 Functional block diagrams and terminal connections. The functional block diagrams and terminal connections shall be as specified on figure 1.

3.2.3 Functional schematic circuits. The functional schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request. The functional schematic shall be as shown on figure 2.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

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TABLE I. Electrical performance characteristics for device types 01 and 02.

Characteristics	Symbol	Conditions $V_S = \pm 15$ V, $R_L = 2k\Omega$, paragraph 3.4 and figure 4 unless otherwise specified		Limits Device 01 Min	Limits Device 01 Max	Limits Device 02 Min	Limits Device 02 Max	
Multiplier accuracy (-x)(-y)	MA_{x-y-}	$V_x = -10$ V	$T_A = 25^\circ C$	-0.5	0.5	-1.0	1.0	%FS
		$V_y = -10$ V	$-55^\circ C \leq T_A \leq 125^\circ C$	-1.0	1.0	-2.0	2.0	
Multiplier accuracy (-x)(+y)	MA_{x-y+}	$V_x = -10$ V	$T_A = 25^\circ C$	-0.5	0.5	-1.0	1.0	"
		$V_y = +10$ V	$-55^\circ C \leq T_A \leq 125^\circ C$	-1.0	1.0	-2.0	2.0	
Multiplier accuracy (+x)(-y)	MA_{x+y-}	$V_x = +10$ V	$T_A = 25^\circ C$	-0.5	0.5	-1.0	1.0	"
		$V_y = -10$ V	$-55^\circ C \leq T_A \leq 125^\circ C$	-1.0	1.0	-2.0	2.0	
Multiplier accuracy (+x)(+y)	MA_{x+y+}	$V_x = +10$ V	$T_A = 25^\circ C$	-0.5	0.5	-1.0	1.0	"
		$V_y = +10$ V	$-55^\circ C \leq T_A \leq 125^\circ C$	-1.0	1.0	-2.0	2.0	
Multiplier accuracy drift	$\frac{\Delta MA_{xy}}{\Delta T}$	$V_x = \pm 10$ V $V_y = \pm 10$ V	$-55^\circ C \leq T_A \leq 125^\circ C$	-0.01	0.01	-0.02	0.02	$\frac{\%FS}{^\circ C}$
Output offset voltage	V_{OIO}	$V_x = V_y = V_z = 0$	$T_A = 25^\circ C$	-15.0	15.0	-30.0	30.0	mV
			$T_A = -55^\circ C$	-39.0	39.0	-70.0	70.0	
			$T_A = 125^\circ C$	-45.0	45.0	-80.0	80.0	
Output offset voltage drift	$\frac{\Delta V_{OIO}}{\Delta T}$		ΔT_A from $-55^\circ C$ to $25^\circ C$ ΔT_A from $25^\circ C$ to $125^\circ C$	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{^\circ C}$
Offset voltage (x)	$V_{IO(x)}$		$T_A = 25^\circ C$	-10.0	10.0	-20.0	20.0	mV
			$T_A = -55^\circ C$	-22.0	22.0	-36.0	36.0	
			$T_A = 125^\circ C$	-25.0	25.0	-40.0	40.0	
Output offset voltage drift (x)	$\frac{\Delta V_{OIO(x)}}{\Delta T}$		ΔT_A from $-55^\circ C$ to $25^\circ C$ ΔT_A from $25^\circ C$ to $125^\circ C$	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{^\circ C}$
Offset voltage (y)	$V_{IO(y)}$		$T_A = 25^\circ C$	-10.0	10.0	-20.0	20.0	mV
			$T_A = -55^\circ C$	-22.0	22.0	-36.0	36.0	
			$T_A = 125^\circ C$	-25.0	25.0	-40.0	40.0	

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TABLE I. Electrical performance characteristics for device types 01 and 02 - Continued.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits				
				Device 01		Device 02		
				Min	Max	Min	Max	
Offset voltage drift (y)	$\frac{V_{IO}(y)}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{°C}$
Offset voltage (z)	$V_{IO}(z)$		T_A = 25°C	-15.0	15.0	-30.0	30.0	mV
			T_A = -55°C	-39.0	39.0	-70.0	70.0	
			T_A = 125°C	-45.0	45.0	-80.0	80.0	
Offset voltage drift (z)	$\frac{V_{IO}(z)}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{°C}$
Input bias current (+)	$+I_{IB}$		T_A = -55°C	-2.5	2.5	-2.5	2.5	µA
			25°C ≤ T_A ≤ 125°C	-2.0	2.0	-2.0	2.0	
Input bias current (-)	$-I_{IB}$		T_A = -55°C	-2.5	2.5	-2.5	2.5	"
			25°C ≤ T_A ≤ 125°C	-2.0	2.0	-2.0	2.0	
Input offset current	I_{IO}		T_A = -55°C	-300.0	300.0	-300.0	300.0	nA
			25°C ≤ T_A ≤ 125°C	-250.0	250.0	-250.0	250.0	
Output short circuit current (+)	$I_{OS}(+)$	R_L = 0; t ≤ 25 ns	-55°C ≤ T_A ≤ 25°C	-40.0		-40.0		mA
			T_A = 125°C	-30.0		-30.0		
Output short circuit current (-)	$I_{OS}(-)$	R_L = 0; t ≤ 25 ns	-55°C ≤ T_A ≤ 25°C		50.0		50.0	"
			T_A = 125°C		38.0		38.0	
Supply current (+)	I_{CC}	R_L = ∞	-55°C ≤ T_A ≤ 125°C		6.5		6.5	"
Supply current (-)	I_{EE}	R_L = ∞	-55°C ≤ T_A ≤ 125°C	-6.5		-6.5		"
Common mode rejection ratio (x)	CMRR(x)	-10 V ≤ V_x ≤ +10 V V_y = 0 V	T_A = -55°C	65.0		54.0		dB
			25°C ≤ T_A ≤ 125°C	70.0		60.0		
Common mode rejection ratio (y)	CMRR(y)	-10 V ≤ V_y ≤ +10 V V_x = 0 V	T_A = -55°C	65.0		54.0		"
			25°C ≤ T_A ≤ 125°C	70.0		60.0		"

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TABLE I. Electrical performance characteristics for device types 01 and 02 - Continued.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits Device 01 Min	Limits Device 01 Max	Limits Device 02 Min	Limits Device 02 Max	
Common mode rejection ratio (z)	CMRR(z)	-10 V ≤Vy ≤+10 V, Vx = +10 V	T_A = -55°C	65.0		54.0		dB
			25°C ≤T_A ≤125°C	70.0		60.0		
Output voltage swing	V_{OP}		-55°C ≤T_A ≤125°C	+11.0		+11.0		V
					-11.0		-11.0	
Power supply rejection ratio (-)	PSRR1	±13.5 V ≤V_S ≤+15 V, Vy = +10 V, Vx = +5 V	-55°C ≤T_A ≤125°C	-10.0	10.0	-10.0	10.0	$\frac{mV}{V}$
Power supply rejection ratio (+)	PSRR2	±15 V ≤V_S ≤+18 V, Vx = +5 V, Vy = +10 V	-55°C ≤T_A ≤125°C	-4.0	4.0	-4.0	4.0	"
Settling time (+)	ts(+)	See figure 5	T_A = 25°C		3.0		3.0	µs
Settling time (-)	ts(-)	See figure 5	T_A = 25°C		3.0		3.0	"
Slew rate (+)	SR(+)		-55°C ≤T_A ≤125°C	18.0		18.0		$\frac{V}{µs}$
Slew rate (-)	SR(-)		-55°C ≤T_A ≤125°C	"		"		"
Feedthrough (x)	FTx		-55°C ≤T_A ≤125°C	-35.0	35.0	-60.0	60.0	mV
Feedthrough (y)	FTy		-55°C ≤T_A ≤125°C	-12.0	12.0	-24.0	24.0	"
Small signal amplitude error (x)	AEx	Vx = 2 V (p-p) Cc = 1000 pf, Vy = +10 V 1% error	T_A 25°C	75.0		75.0		kHz
Small signal amplitude error (y)	AEy	Vy = 2 V(p-p) Cc = 1000 pf, Vx = +10 V 1% error	T_A 25°C	"		"		"
Nonlinearity (x)	NLx		-55°C ≤T_A ≤125°C	-0.3	0.3	-0.5	0.5	%FS
Nonlinearity (y)	NLy		-55°C ≤T_A ≤125°C	-0.15	0.15	-0.2	0.2	"
Wideband noise	NI(BB)		T_A = 25°C		1.5		1.5	mV rms

MIL-M-38510/139B

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TABLE I. Electrical performance characteristics for device type 03.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits Min	Limits Max	Unit
Multiplier accuracy (-x)(-y)	MAx-y-	Vx = -10 V	T_A = 25°C	-1.0	1.0	%FS
		Vy = -10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (-x)(+y)	MAx-y+	Vx = -10 V	T_A = 25°C	-1.0	1.0	"
		Vy = +10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (+x)(-y)	MAx+y-	Vx = +10 V	T_A = 25°C	-1.0	1.0	"
		Vy = -10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (+x) (+y)	MAx+y+	Vx = +10 V	T_A = 25°C	-1.0	1.0	"
		Vy = +10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy drift	$\frac{MA\ xy}{\Delta T}$	Vx = ±10 V Vy = ±10 V	-55°C ≤ T_A ≤ 125°C	-0.04	0.04	$\frac{\%FS}{°C}$
Output offset voltage	V_{OIO}	Vx = Vy = Vz = 0	T_A = 25°C	-30.0	30.0	mV
			T_A = -55°C	-190.0	190.0	
			T_A = 125°C	-230.0	230.0	
Output offset voltage drift	$\frac{V_{OIO}}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-2.0	2.0	$\frac{mV}{°C}$
Offset voltage (x)	$V_{IO}(x)$		T_A = 25°C	-50.0	50.0	mV
			T_A = -55°C	-100.0	100.0	
			T_A = 125°C	-100.0	100.0	
Output offset voltage drift (x)	$\frac{V_{OIO}(x)}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-800.0	800.0	$\frac{\mu V}{°C}$
Offset voltage (y)	$V_{IO}(y)$		T_A = 25°C	-50.0	50.0	mV
			T_A = -55°C	-100.0	100.0	
			T_A = 125°C	-100.0	100.0	

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TABLE I. Electrical performance characteristics for device type 03 - Continued.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits Min	Limits Max	Unit
Offset voltage drift (y)	$\frac{\Delta V_{IO(y)}}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-800.0	800.0	$\frac{\mu V}{°C}$
Input bias current (z)	$I_{IB(z)}$		-55°C ≤ T_A ≤ 125°C	-12.0	12.0	µA
Input bias current (+)	$+I_{IB}$		T_A = -55°C	-4.0	4.0	"
			25°C ≤ T_A ≤ 125°C	-2.0	2.0	
Input bias current (-)	$-I_{IB}$		T_A = -55°C	-4.0	4.0	"
			25°C ≤ T_A ≤ 125°C	-2.0	2.0	
Input offset current	I_{IO}		T_A = -55°C	-400.0	400.0	nA
			25°C ≤ T_A ≤ 125°C	-200.0	200.0	
Output short circuit current (+)	$I_{OS(+)}$	R_L = 0	-55°C ≤ T_A ≤ 25°C	-45.0		mA
			T_A = 125°C	-30.0		
Output short circuit current (-)	$I_{OS(-)}$	R_L = 0	-55°C ≤ T_A ≤ 25°C		30.0	"
			T_A = 125°C		"	
Supply current (+)	I_{CC}	R_L = "	-55°C ≤ T_A ≤ 125°C		6.5	"
Supply current (-)	I_{EE}	R_L = "	-55°C ≤ T_A ≤ 125°C	-6.5		"
Common mode rejection ratio (x)	CMRR(x)	-10 V ≤ Vx ≤ +10 V Vy = +10 V	-55°C ≤ T_A ≤ 125°C	50.0		dB
Common mode rejection ratio (y)	CMRR(y)	-10 V ≤ Vy ≤ +10 V Vx = +10 V	T_A = -55°C 25°C ≤ T_A ≤ 125°C	"		"

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TABLE I. Electrical performance characteristics for device type 03 - Continued.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits		Unit
				Min	Max	
Common mode rejection ratio (y)	CMRR(y)	-10 V ≤Vy ≤+10 V Vx = +10 V	T_A = -55°C 25°C ≤T_A ≤125°C	50.0		dB
Output voltage swing	V_{OP}		-55°C ≤T_A ≤125°C	+10.0		V
					-10.0	
Power supply rejection ratio (-)	PSRR1	±13.5 V ≤V_S ≤±15 V, Vy = +10 V, Vx = +5 V	-55°C ≤T_A ≤125°C	-50.0	50.0	$\frac{mV}{V}$
Power supply rejection ratio (+)	PSRR2	±15 V ≤V_S ≤±18 V, Vx = -5 V, Vy = +10 V	-55°C ≤T_A ≤125°C	"	"	"
Settling time (+)	ts(+)	See figure 5	T_A = 25°C		2.0	µs
Settling time (-)	ts(-)	See figure 5	T_A = 25°C		"	"
Slew rate (+)	SR(+)		-55°C ≤T_A ≤125°C	25.0		$\frac{V}{\mu s}$
Slew rate (-)	SR(-)		-55°C ≤T_A ≤125°C	35.0		"
Feedthrough (x)	FTx		-55°C ≤T_A ≤125°C	-100.0	100.0	mV
Feedthrough (y)	FTy		-55°C ≤T_A ≤125°C	-80.0	80.0	"
Small signal amplitude error (x)	AEx	Vx = 2 V(p-p) Cc = 1000 pF, Vy = +10 V 1% error	T_A 25°C	70.0		kHz
Small signal amplitude error (y)	AEy	Vy = 2 V(p-p) Cc = 1000 pF, Vx = +10 V 1% error	T_A 25°C	"		"
Nonlinearity (x)	NLx		-55°C ≤T_A ≤125°C	-0.6	0.6	%FS
Nonlinearity (y)	NLy		-55°C ≤T_A ≤125°C	-0.4	0.4	"
Wideband noise	NI(BB)		T_A = 25°C		3.0	mV rms

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TABLE I. Electrical performance characteristics for device type 04.

Characteristics	Symbol	Conditions V_S = ±15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits Min	Limits Max	Unit
Multiplier accuracy (-x)(-y)	MAx-y-	V_x = -10 V	T_A = 25°C	-1.0	1.0	%FS
		V_y = -10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (-x)(+y)	MAx-y+	V_x = -10 V	T_A = 25°C	-1.0	1.0	"
		V_y = +10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (+x)(-y)	MAx+y-	V_x = +10 V	T_A = 25°C	-1.0	1.0	"
		V_y = -10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy (+x)(+y)	MAx+y+	V_x = +10 V	T_A = 25°C	-1.0	1.0	"
		V_y = +10 V	-55°C ≤ T_A ≤ 125°C	-4.0	4.0	
Multiplier accuracy drift	$MA_{\frac{xy}{\Delta T}}$	V_x = ±10 V V_y = ±10 V	-55°C ≤ T_A ≤ 125°C	-0.04	0.04	$\frac{\%FS}{°C}$
Output offset voltage	V_{O10}	V_x = V_y = V_z = 0	T_A = 25°C	-30.0	30.0	mV
			T_A = -55°C	-100.0	100.0	
			T_A = 125°C	"	"	
Output offset voltage drift	$\frac{V_{O10}}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-1.0	1.0	$\frac{mV}{°C}$
Offset voltage (x)	$V_{IO}(x)$		T_A = 25°C	-40.0	40.0	mV
			T_A = -55°C	-90.0	90.0	
			T_A = 125°C	"	"	
Output offset voltage drift (x)	$\frac{V_{O10}(x)}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-800.0	800.0	$\frac{\mu V}{°C}$
Offset voltage (y)	$V_{IO}(y)$		T_A = 25°C	-50.0	50.0	mV
			T_A = -55°C	-100.0	100.0	
			T_A = 125°C	"	"	

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TABLE I. Electrical performance characteristics for device type 04 - Continued.

Characteristics	Symbol	Conditions $V_S = \pm 15$ V, $R_L = 2k\Omega$, paragraph 3.4 and figure 4 unless otherwise specified		Limits		Unit
				Min	Max	
Offset voltage drift (y)	$\frac{\Delta V_{IO}(y)}{\Delta T}$		ΔT_A from -55°C to 25°C ΔT_A from 25°C to 125°C	-800.0	800.0	$\frac{\mu V}{°C}$
Input bias current (z)	$I_{IB}(z)$		-55°C ≤ T_A ≤ 125°C	---	6.0	µA
Input bias current (+)	$+I_{IB}$		T_A = -55°C	-6.0	"	"
			25°C ≤T_A ≤125°C	-2.5	2.5	
Input offset current (-)	$-I_{IB}$		T_A = -55°C	-6.0	6.0	"
			25°C ≤T_A ≤125°C	-2.5	2.5	
Input offset current	I_{IO}		T_A = 25°C	-200.0	200.0	nA
			-55°C ≤T_A ≤125°C	-400.0	400.0	
Output short circuit current (+)	$I_{OS}(+)$	R_L = 0	-55°C ≤T_A ≤25°C	-30.0		mA
			T_A = 125°C	"		
Output short circuit current (-)	$I_{OS}(-)$	R_L = 0	-55°C ≤T_A ≤25°C		30.0	"
			T_A = 125°C		"	
Supply current (+)	I_{CC}	R_L = ∞	-55°C ≤T_A ≤125°C		6.0	"
Supply current (-)	I_{EE}	R_L = ∞	-55°C ≤T_A ≤125°C	-6.0		"
Common mode rejection ratio (x)	CMRR(x)	-6 V ≤Vx ≤+10 V Vy = ∓10 V	-55°C ≤ T_A ≤125°C	50.0		dB
Common mode rejection ratio (y)	CMRR(y)	-6 V ≤Vy ≤+10 V Vx = ∓10 V	-55°C ≤ T_A ≤ 125°C	"		"

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TABLE I. Electrical performance characteristics for device type 04 - Continued.

Characteristics	Symbol	Conditions V_S = +15 V, R_L = 2kΩ, paragraph 3.4 and figure 4 unless otherwise specified		Limits		Unit
				Min	Max	
Output voltage swing	V_{OP}		-55°C ≤ T_A ≤ 125°C	+10.0		V
					-10.0	
Power supply rejection ratio (-)	PSRR1	±13.5 V ≤ V_S ≤ +15 V, V_y = +10 V, V_x = +5 V	-55°C ≤ T_A ≤ 125°C	-50.0	50.0	$\frac{mV}{V}$
Power supply rejection ratio (+)	PSRR2	±15 V ≤ V_S ≤ ±18 V, V_x = +5 V, V_y = +10 V	-55°C ≤ T_A ≤ 125°C	-40.0	40.0	"
Settling time (+)	ts(+)	See figure 5	T_A = 25°C		2.0	µs
Settling time (-)	ts(-)	See figure 5	T_A = 25°C		"	"
Slew rate (+)	SR(+)		-55°C ≤ T_A ≤ 125°C	20.0		$\frac{V}{\mu s}$
Slew rate (-)	SR(-)		-55°C ≤ T_A ≤ 125°C	"		"
Feedthrough (x)	FTx		-55°C ≤ T_A ≤ 125°C	-50.0	50.0	mV
Feedthrough (y)	FTy		-55°C ≤ T_A ≤ 125°C	-40.0	40.0	"
Small signal amplitude error (x)	AEx	V_x = 2 V (p-p) Cc = 1000 pF, V_y = +10 V 1% error	T_A = 25°C	70.0		kHz
Small signal amplitude error (y)	AEy	V_y = 2 V(p-p) Cc = 1000 pF, V_x = +10 V 1% error	T_A = 25°C	"		"
Nonlinearity (x)	NLx		-55°C ≤ T_A ≤ 125°C	-0.6	0.6	%FS
Nonlinearity (y)	NLy		-55°C ≤ T_A ≤ 125°C	-0.4	0.4	"
Wideband noise	NI(BB)		T_A = 25°C		3.0	mV rms

MIL-M-38510/139B

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3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics of table I apply over the full recommended ambient temperature range. Unless otherwise specified, source resistance is 50 ohms for all tests.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1, 2, 3 1/	1, 2, 3 1/
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7, 9	1, 2, 3, 4, 5, 6, 7, 9
Group C end-point and group B, class S, electrical parameters (method 5005)	1, 2, 3 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	---
Group D end-point electrical parameters (method 5005)	1, 2, 3	1

1/ PDA applies to subgroup 1 (see 4.2c).

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 49 or I (see MIL-M-38510, appendix E).

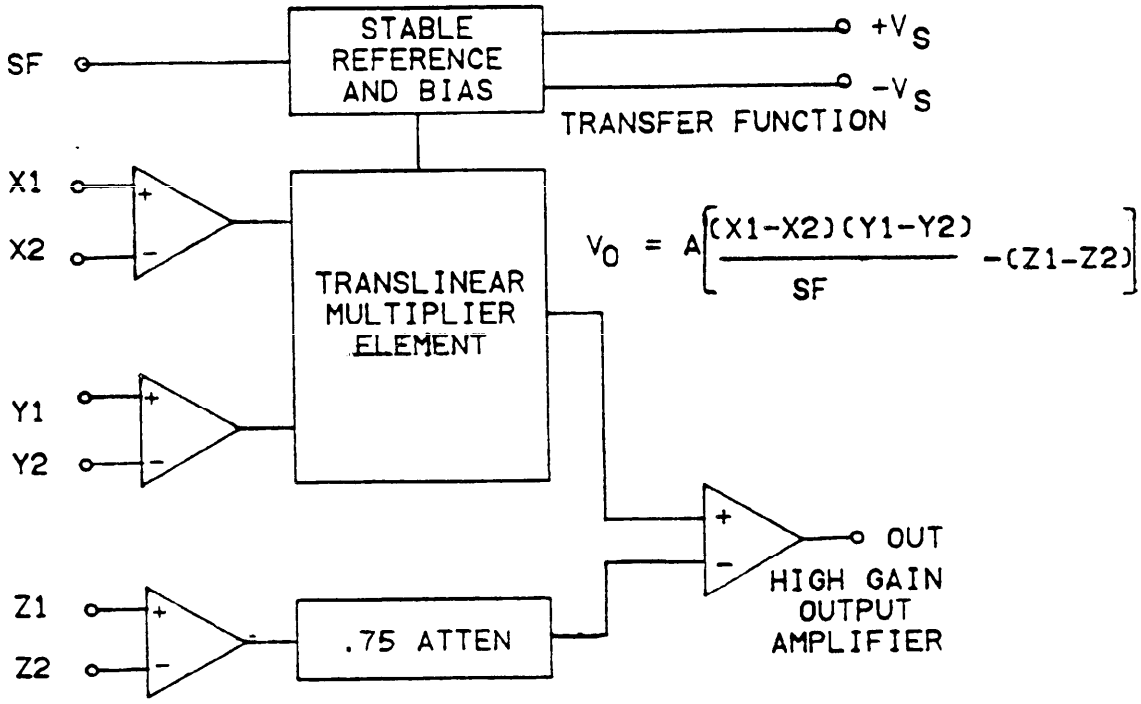
4. QUALITY ASSURANCE PROVISIONS

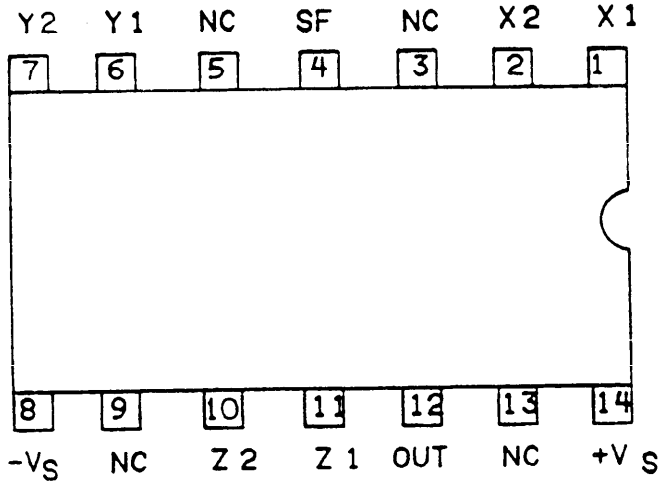
4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, except as modified herein.

Text continues on page 60.

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Device types 01 and 02





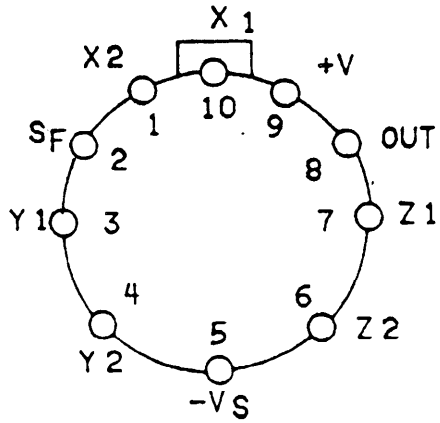
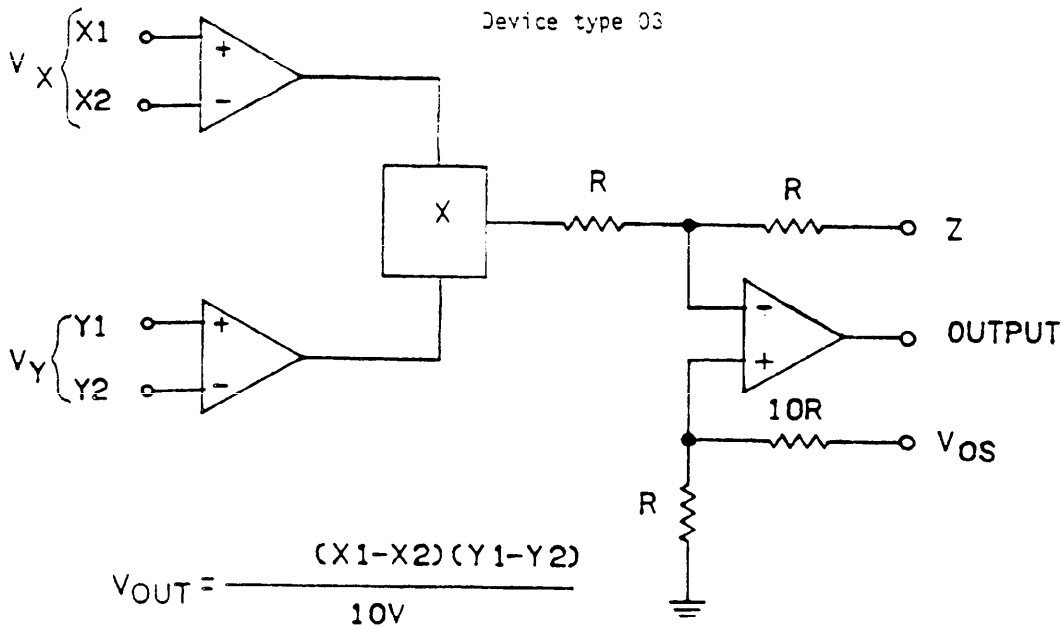
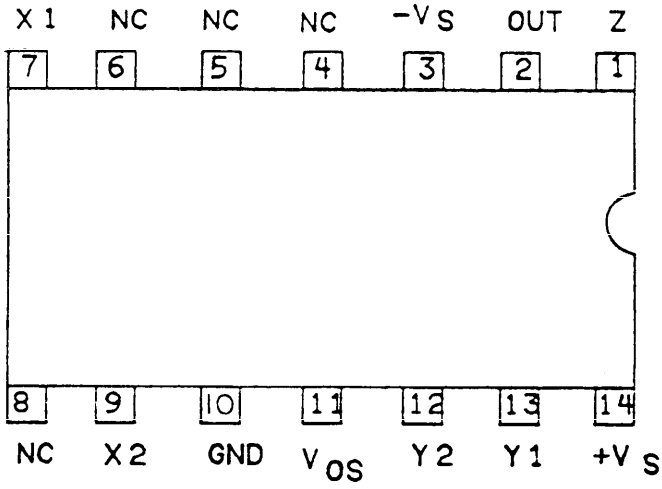


FIGURE 1. Functional block diagrams and terminal connections.

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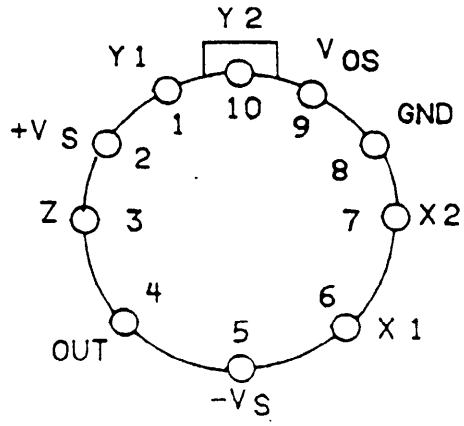
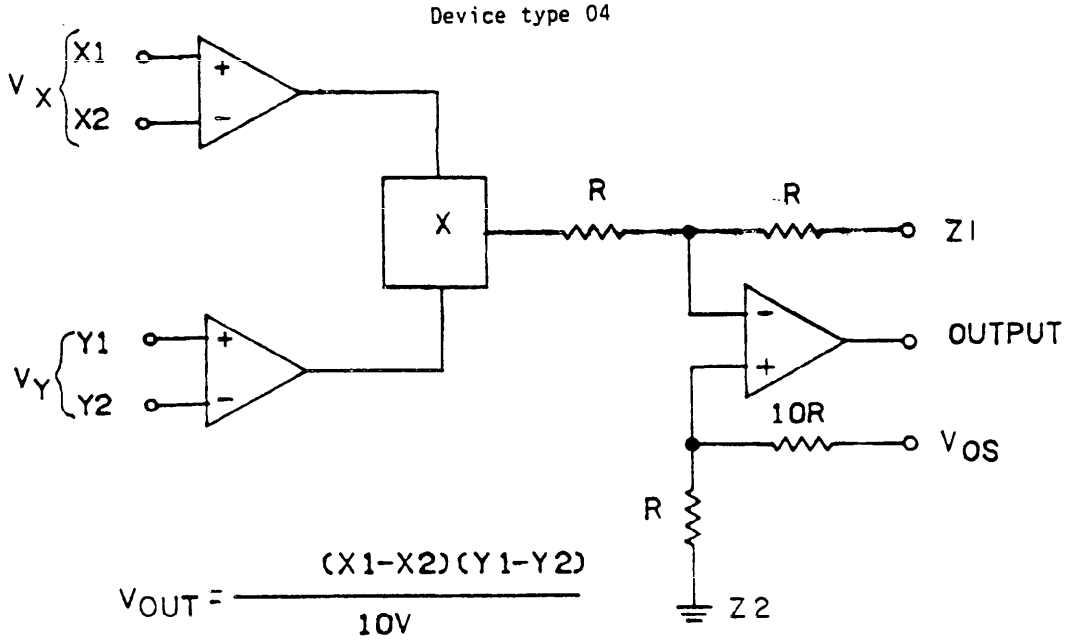


FIGURE 1. Functional block diagrams and terminal connections - Continued.

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Device type 04



$$V_{OUT} = \frac{(X1-X2)(Y1-Y2)}{10V}$$

WITH Z_1 TIED TO OUTPUT

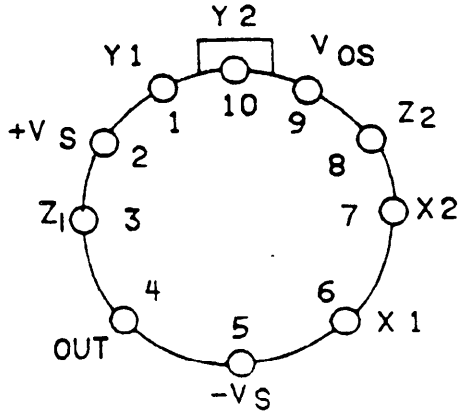


FIGURE 1. Functional block diagrams and terminal connections - Continued.

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Device type 01-02

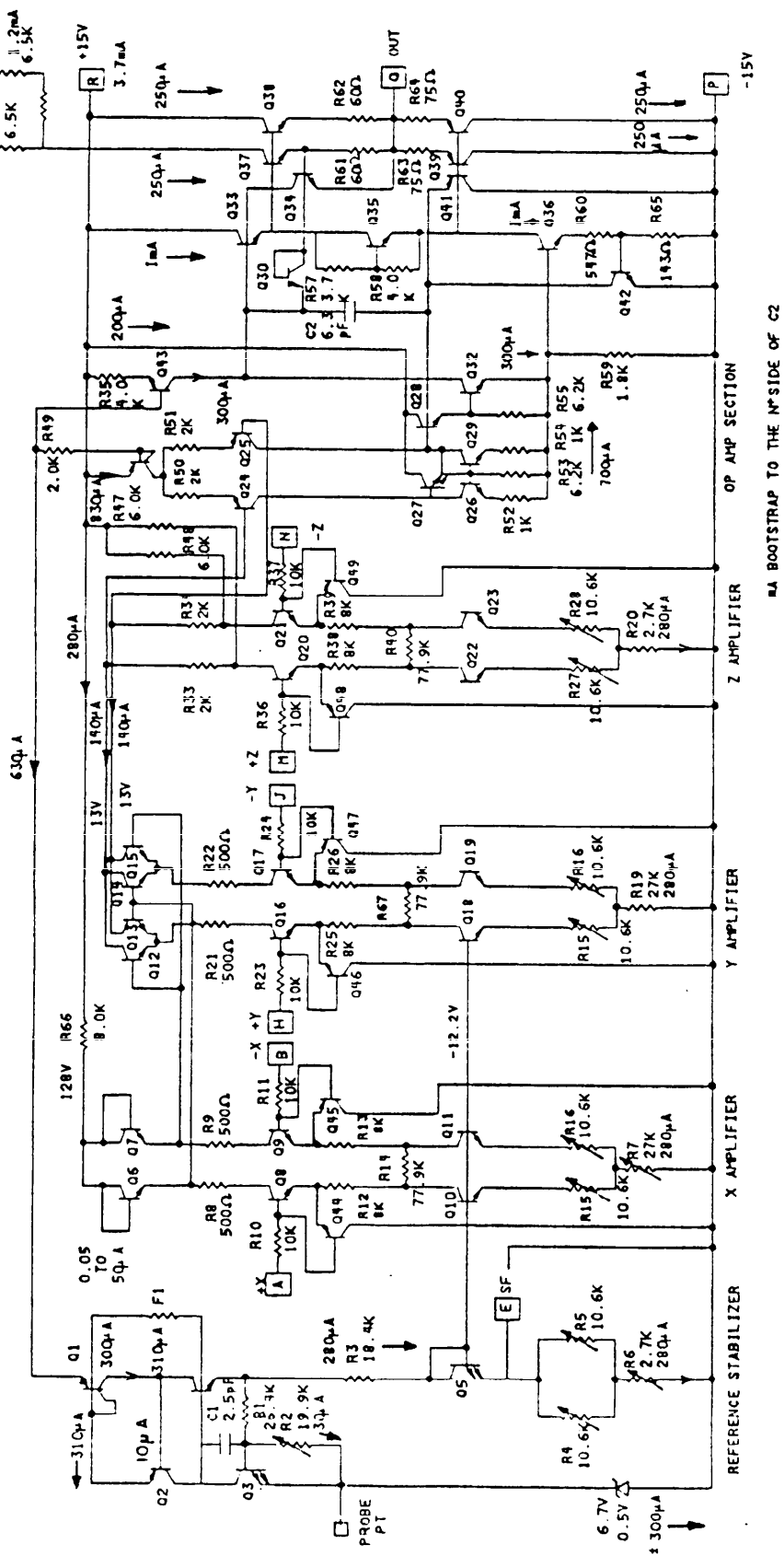


FIGURE 2. Functional schematic circuits.

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Device types 01 and 02

FUNCTION	TRIM CHARACTERISTIC	R#
Gain	Decrease (coarse) Decrease (fine) Increase (coarse)	R6 R4, R5 R7, R3
Xos	-Ref. to +XIN +Ref. to +XIN	R15 R16
Yos	-Ref. to +YIN +Ref. to +YIN	R17 R18
Zos	-Ref. to +ZIN +Ref. to +ZIN	R27 R28
TC	V_{Z1} > 6.7 V V_{Z1} < 6.7 V	R1 R2

NOTES:
1. All resistors are ±20% tolerance and are thin film, 1 kΩ/sq. except R28, R29, and R30.
2. Pairs matched to 1 mV, minimum temperature difference, (Q1 and Q2), (Q7 and Q8), and (Q14 and Q15).
3. Resistor rations matched to 2% (R3 and R4), (R27 and R13).
4. Q28 = Buried zener, 6.5 V ±0.5 V at 240 µA.
5. Resistor values are ±20% and are expressed in ohms, except where otherwise specified.

FIGURE 2. Functional schematic circuits - Continued.

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Device type 03

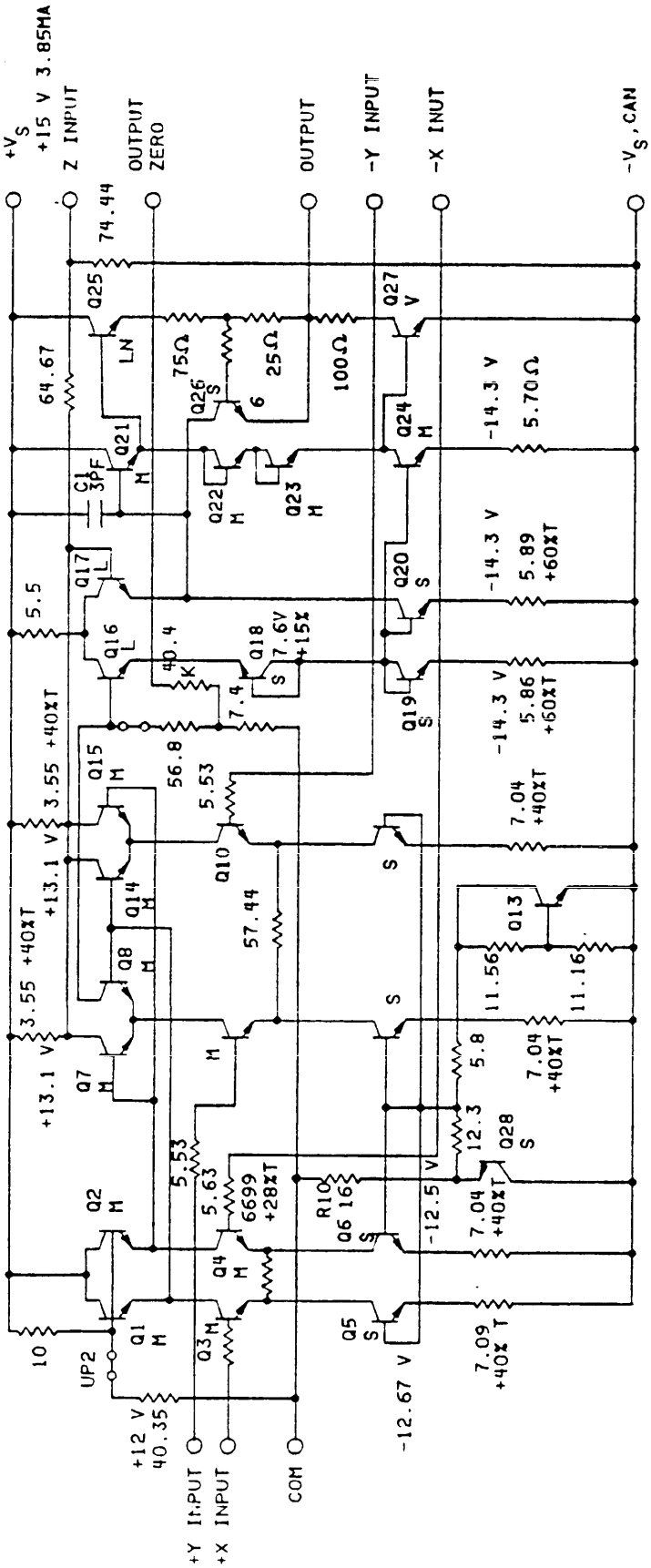
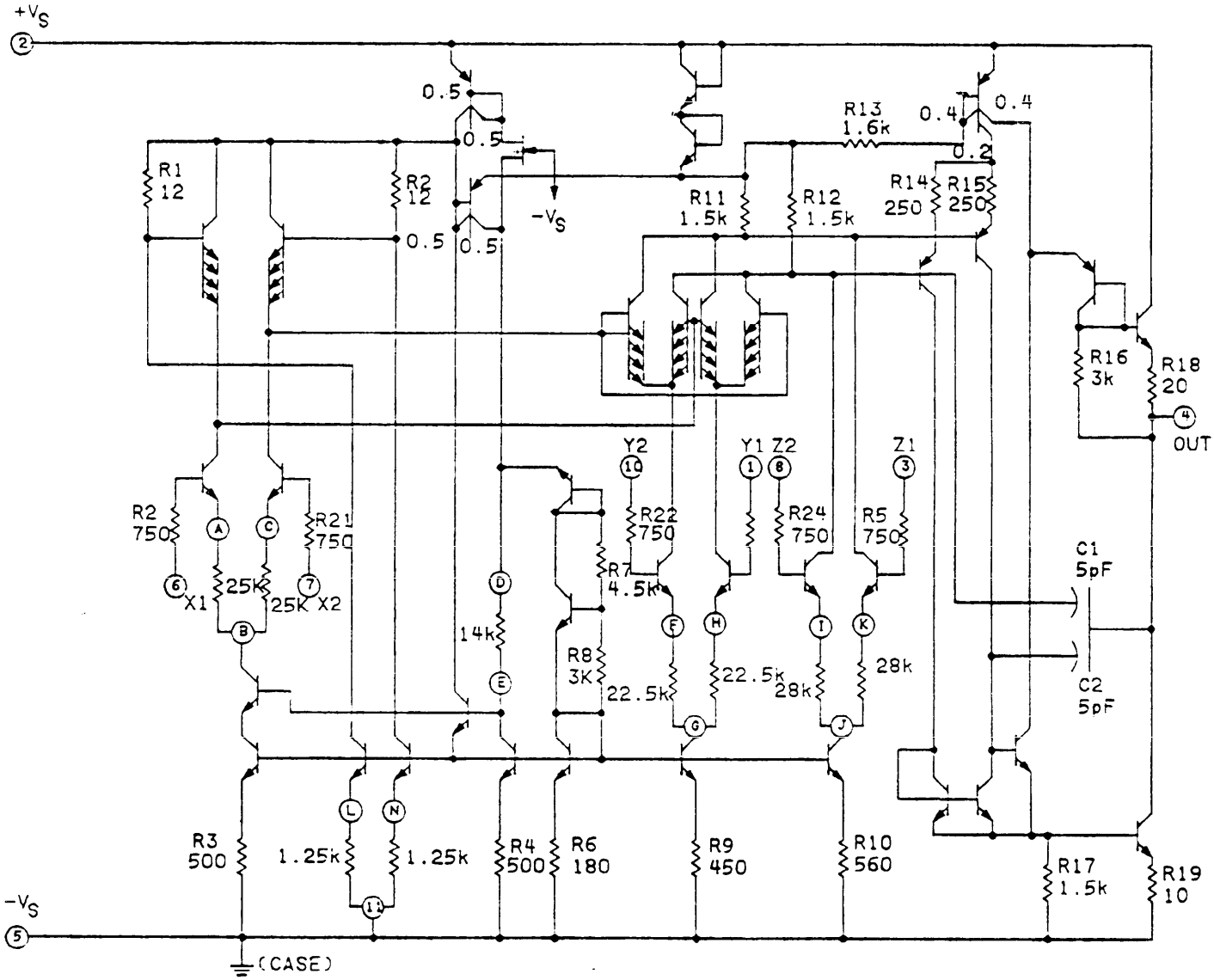


FIGURE 2. Functional schematic circuits - Continued.

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Device type 04



Note: All resistances are nominal and in ohms

FIGURE 2. Functional schematic circuits - Continued.

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Device types 01 and 02

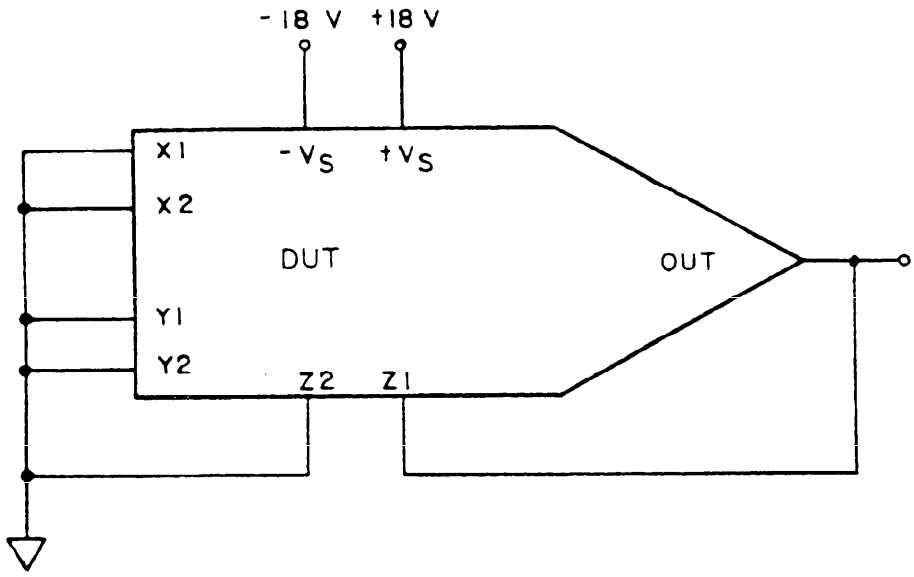


FIGURE 3. Burn-in and steady state life test circuit.

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Device type 03

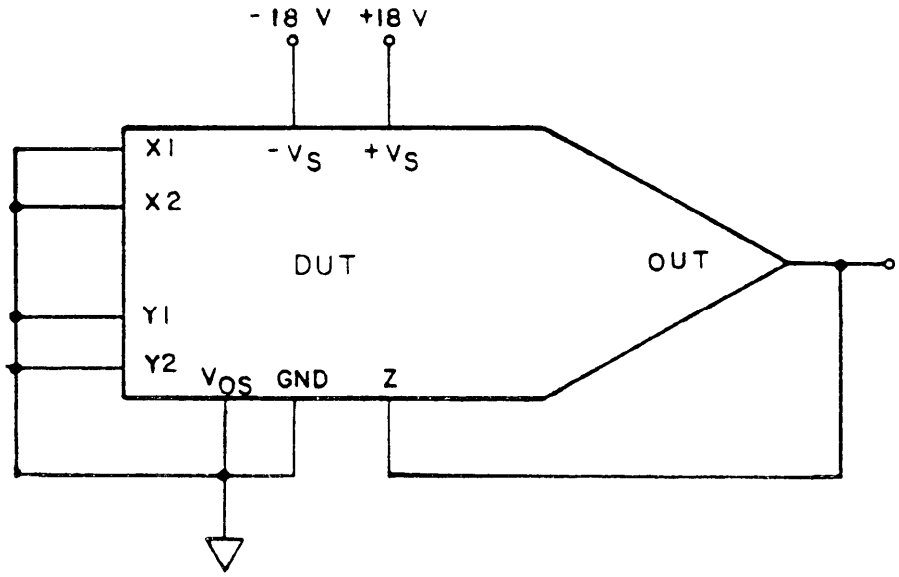


FIGURE 3. Burn-in and steady state life test circuit - Continued.

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Device type 04

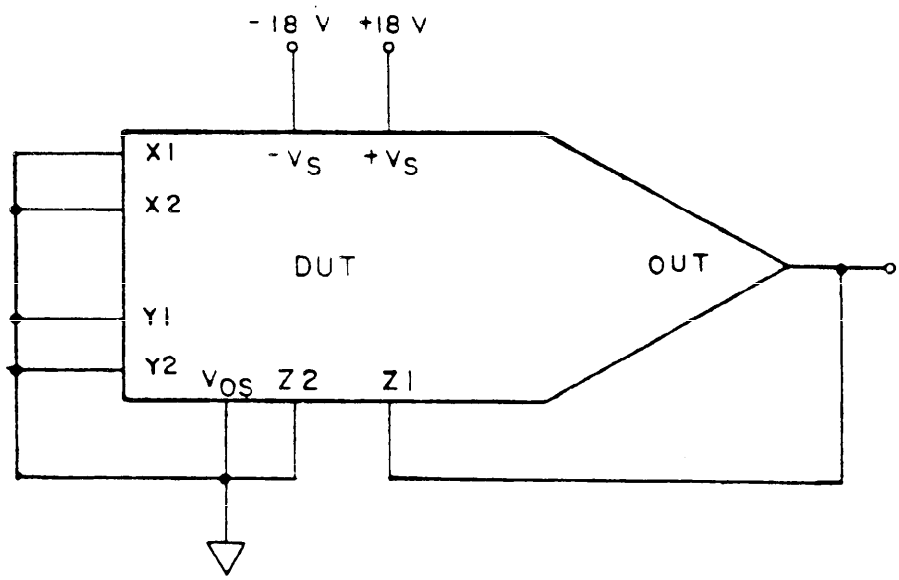


FIGURE 3. Burn-in and steady state life test circuit - Continued.

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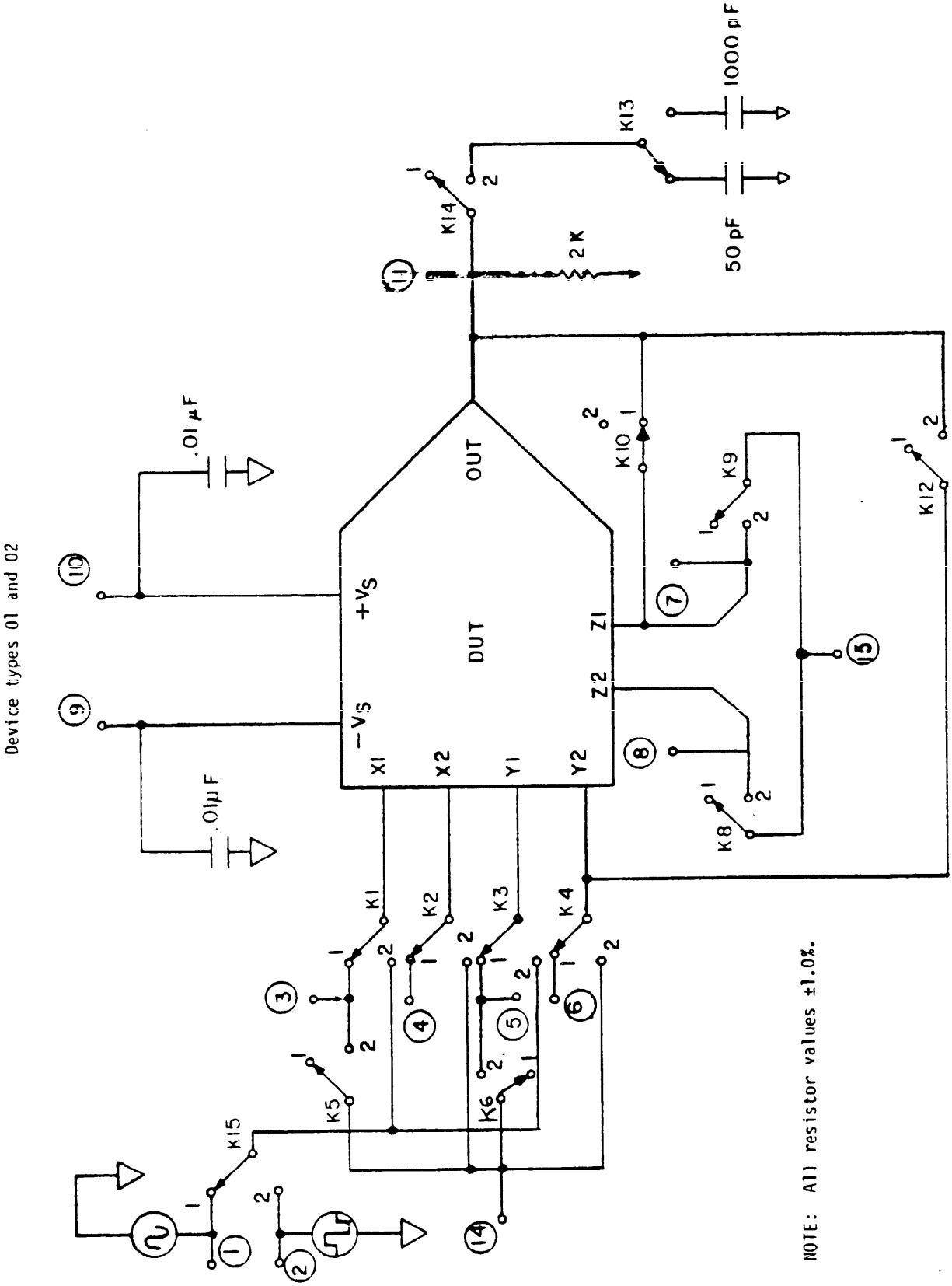
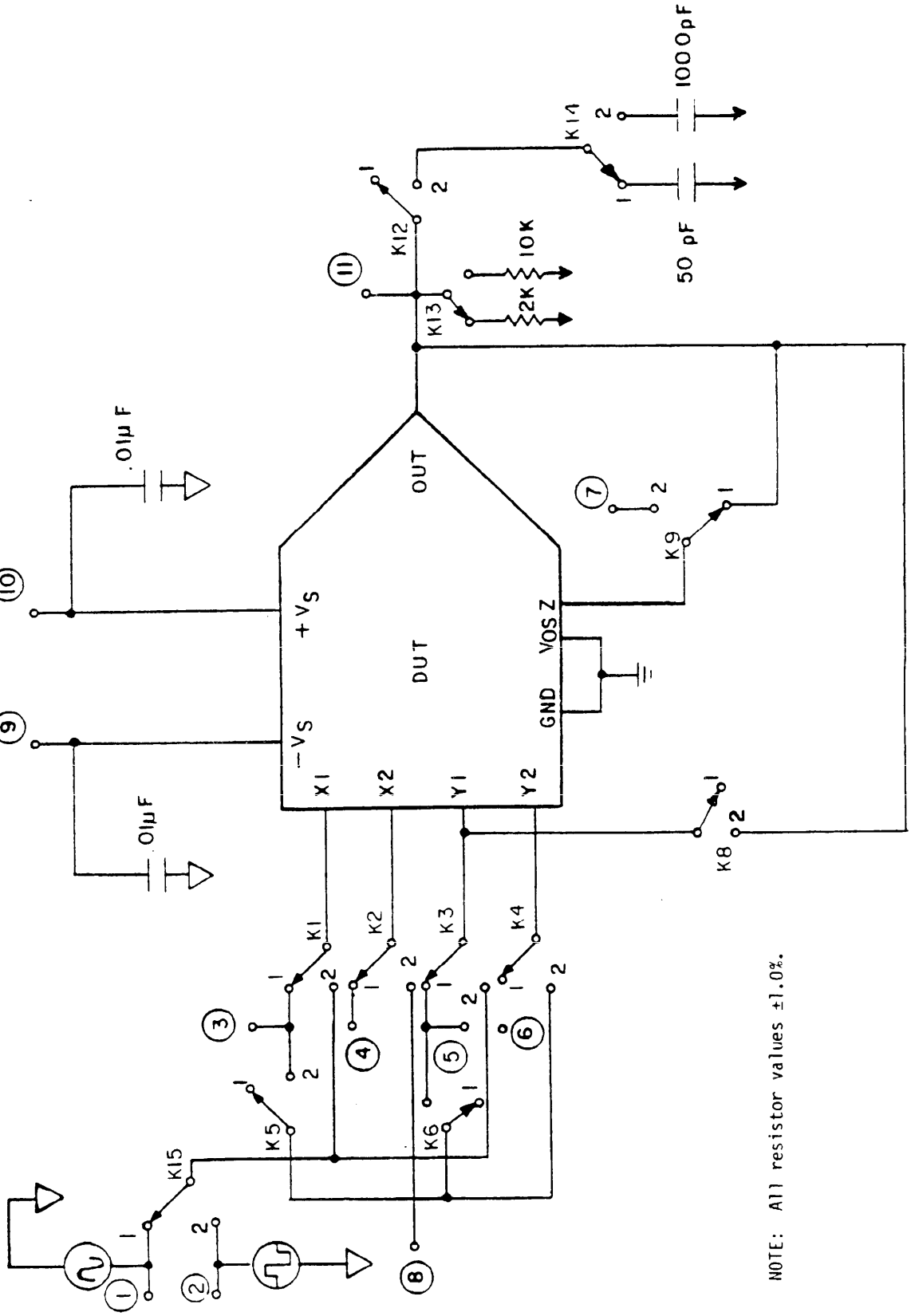


FIGURE 4. Static and dynamic test circuit.

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NOTE: All resistor values ±1.0%.

FIGURE 4. Static and dynamic test circuit - Continued.

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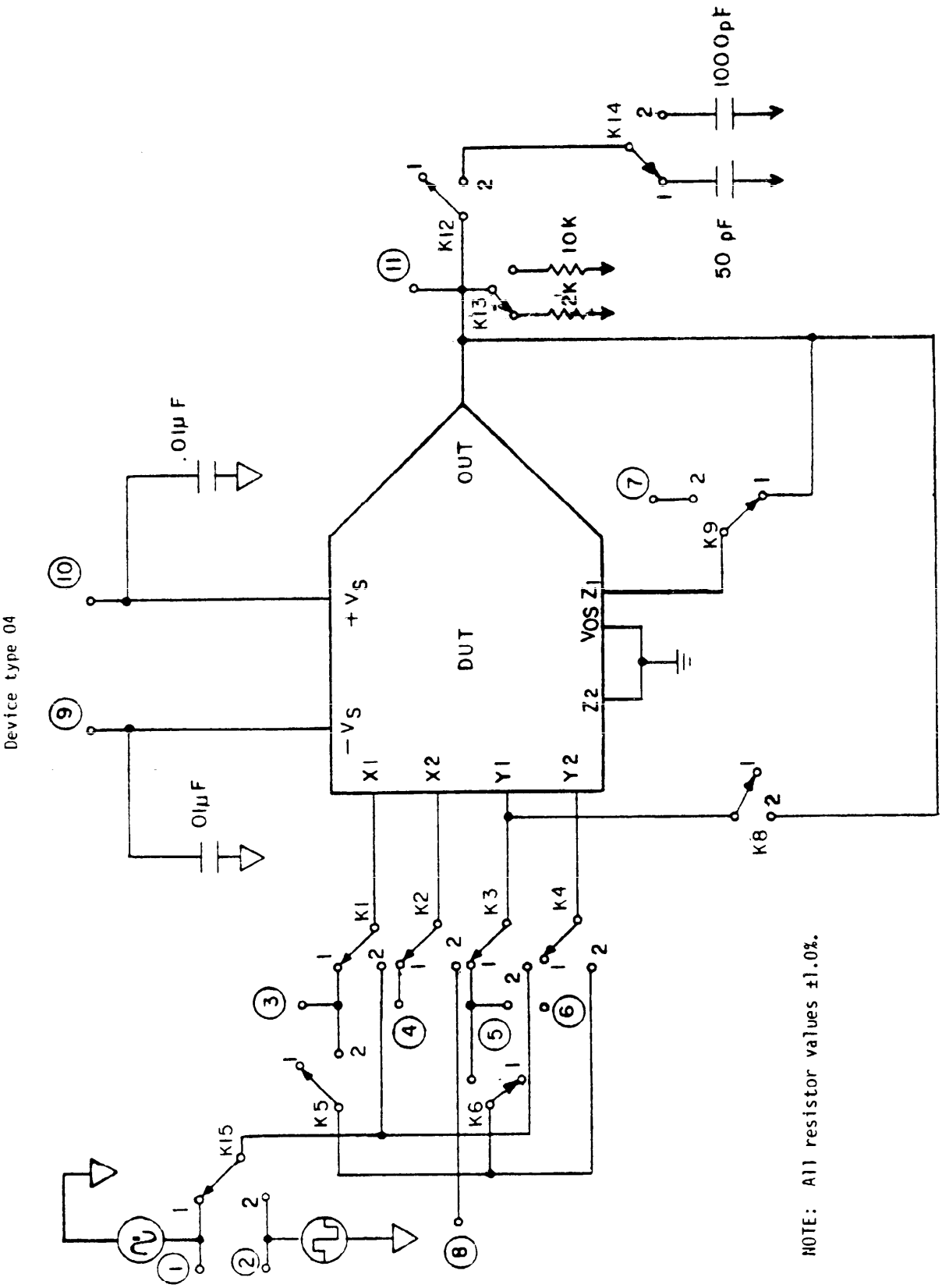


FIGURE 4. Static and dynamic test circuit - Continued.

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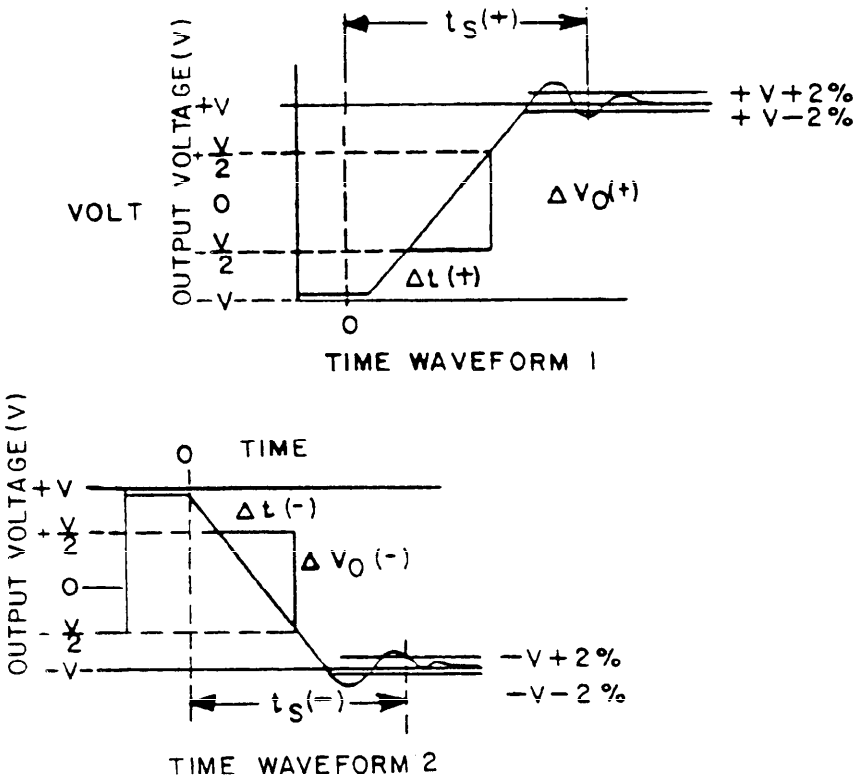
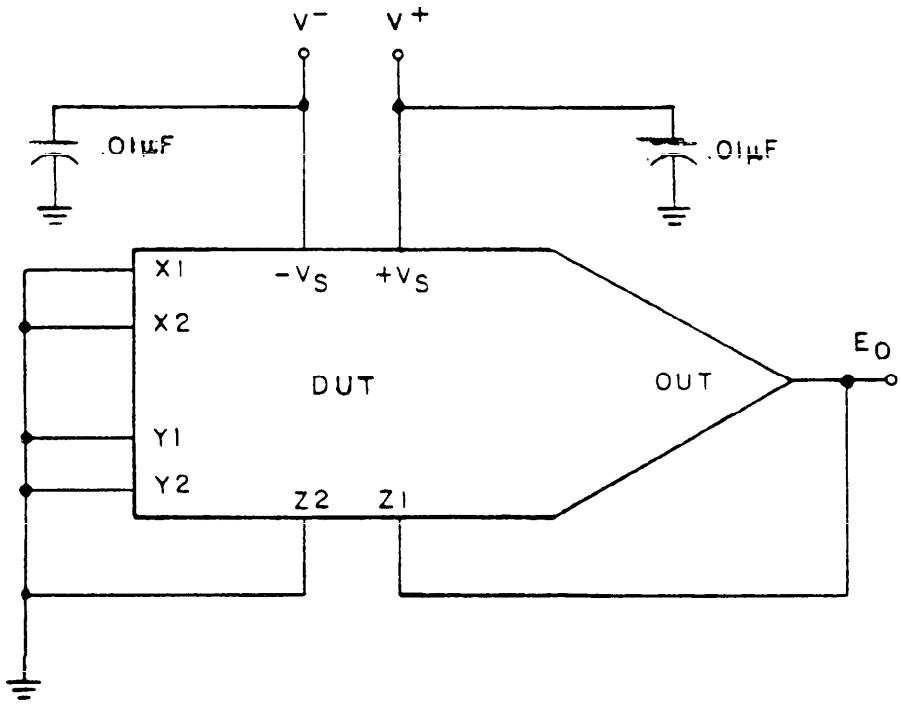


FIGURE 5. Transient response waveforms.

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Devices 01 and 02

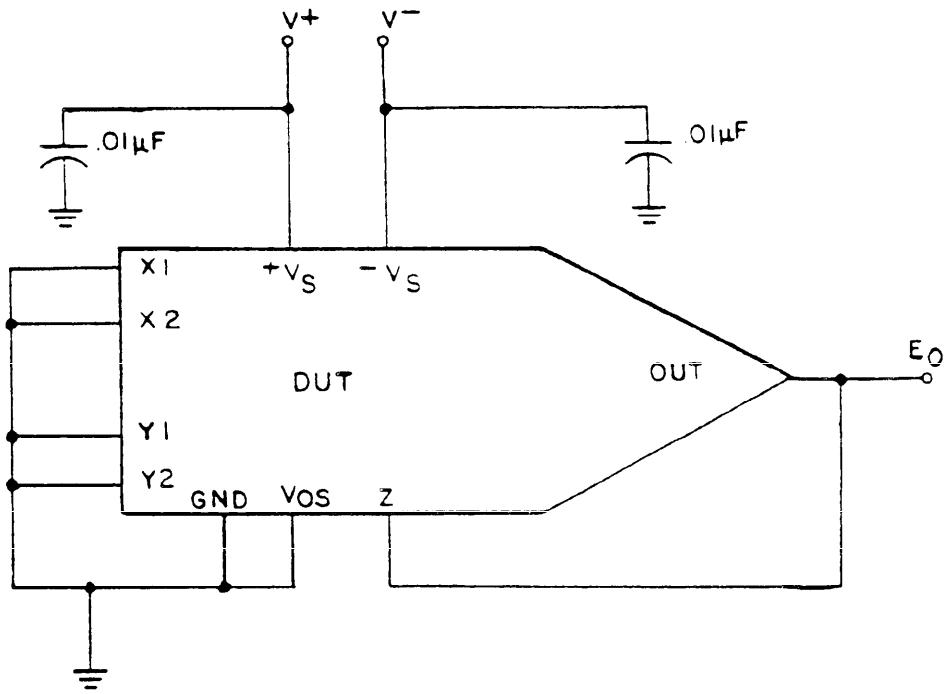


NOTE: E_0 is measured using a rms voltmeter with a bandwidth of 10 Hz to 100 kHz.

FIGURE 6. Noise test circuit.

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Device type 03

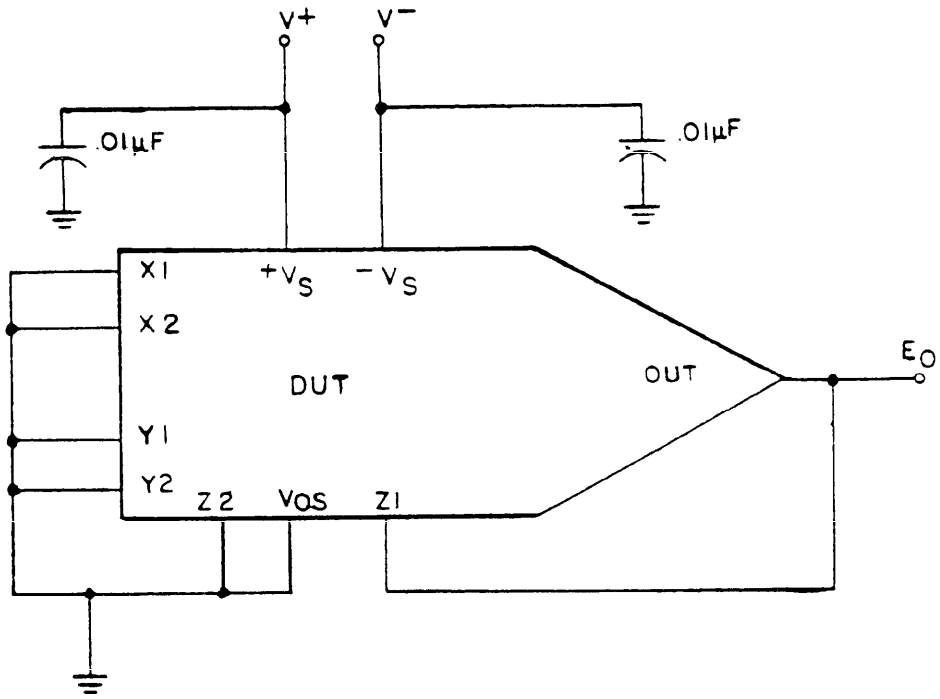


NOTE: E_0 is measured using a rms voltmeter with a bandwidth of 10 Hz to 100 kHz.

FIGURE 6. Noise test circuit - Continued.

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Device type 04

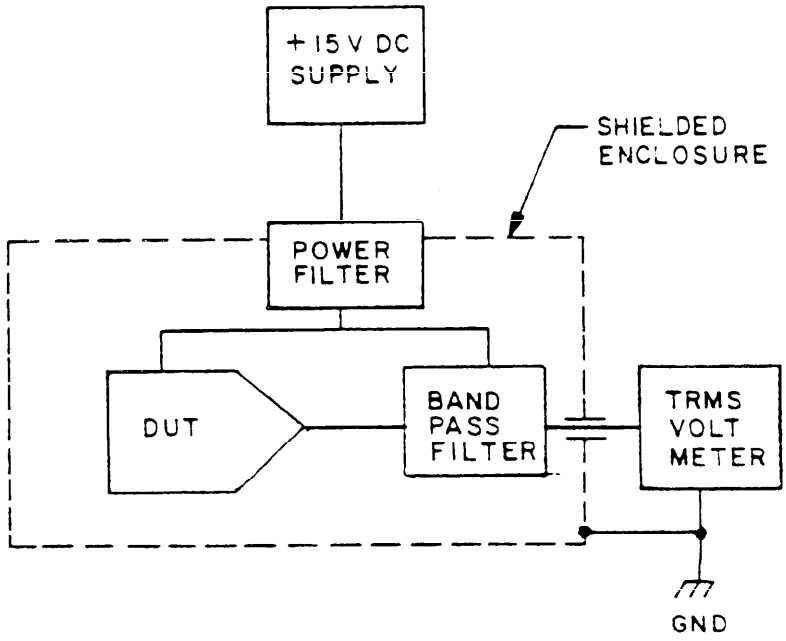


NOTE: E_0 is measured using a rms voltmeter with a bandwidth of 10 Hz to 100 kHz.

FIGURE 6. Noise test circuit - Continued.

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Device types 01, 02, 03 and 04



NOTES:
1. Bandpass filter:
   Passband gain = 40 dB ±0.5 dB.
   Fc low = 10 Hz ±1 Hz.
   Fc high = 100 kHz ±10 kHz.
   Attenuation slope = 12 dB/Oct.
2. V_{NOISE} = Reading/100
3. Voltmeter: HP3400A or equivalent.
4. ±15 V dc power supply HP6235A or equivalent.

FIGURE 6. Noise measurement block diagram - Continued.

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TABLE III. Group A Inspection for device types 01 and 02.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
1 T_A = +25°C	MAx-y-	1	---	---	-10 V	0 V	-10 V	0 V	---	0 V	-15 V	+15 V		11	E1	V	$MAx\text{-}y\text{-} = \frac{(E1-10)}{10} \times 100$	-0.5	0.5	-1.0	1.0	%FS
	MAx-y+	2	---	---	"	"	+10 V	"	---	"	"	"		"	E2	"	$MAx\text{-}y^{+} = \frac{(E2+10)}{10} \times 100$	"	"	"	"	"
	MAx+y-	3	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E3	"	$MAx^{+}y\text{-} = \frac{(E3+10)}{10} \times 100$	"	"	"	"	"
	MAx+y+	4	---	---	"	"	+10 V	"	---	"	"	"		"	E4	"	$MAx^{+}y^{+} = \frac{(E4-10)}{10} \times 100$	"	"	"	"	"
	V_{OIO}	5	---	---	0 V	"	0 V	"	---	"	"	"		"	E5	mV	$V_{OIO} = E5$	-15.0	15.0	-30.0	30.0	mV
	$V_{IO}(x)$	6	---	---	"	"	-10 V	"	---	"	"	"		"	E6	"	$V_{IO}(x) = \frac{(E6-E7)}{2}$	-10.0	10.0	-20.0	20.0	"
	$V_{IO}(x)$	7	---	---	"	"	+10 V	"	---	"	"	"		"	E7	"						
	$V_{IO}(y)$	8	---	---	-10 V	"	0 V	"	---	"	"	"		"	E8	"	$V_{IO}(y) = \frac{(E8-E9)}{2}$	"	"	"	"	"
	$V_{IO}(y)$	9	---	---	+10 V	"	"	"	---	"	"	"		"	E9	"						
	$V_{IO}(z)$	10	---	---	+1 V	"	"	---	0 V	"	"	"	K10, K12	"	E10	"	$V_{IO}(z) = \frac{(E10)}{10}$	-15.0	15.0	-30.0	30.0	"
	$+I_{IB}(x)$	11	---	---	---	"	"	0 V	"	"	"	"	K5	14	I1	µA		-2.0	2.0	-2.0	2.0	µA
	$+I_{IB}(y)$	12	---	---	0 V	"	---	"	"	"	"	"	K6	"	I2	"		"	"	"	"	"
	$+I_{IB}(z)$	13	---	---	"	"	0 V	"	---	"	"	"	K9, K10, K12	15	I3	"		"	"	"	"	"
	$-I_{IB}(x)$	14	---	---	"	---	"	"	0 V	"	"	"	K2	14	I4	"		"	"	"	"	"
	$-I_{IB}(y)$	15	---	---	"	0 V	"	---	"	"	"	"	K4	"	I5	"		"	"	"	"	"
	$-I_{IB}(z)$	16	---	---	"	"	"	0 V	"	---	"	"	K8	15	I6	"		"	"	"	"	"
	$I_{IO}(x)$	17	---	---	"	"	"	"	"	0 V	"	"		---	---	---	$I_{IO}(x) = I1\text{-}I4$	-250.0	250.0	-250.0	250.0	nA
	$I_{IO}(y)$	18	---	---	"	"	"	"	"	---	"	"		---	---	---	$I_{IO}(y) = I2\text{-}I5$	"	"	"	"	"

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TABLE III. Group A inspection fordevice types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
1 T_A = +25°C	$I_{IO}(z)$	19	---	---	0 V	0 V	0 V	0 V	0 V	0 V	-15 V	+15 V		---	---	---	$I_{IO}(z) = I3-I6$	-250.0	250.0	-250.0	250.0	nA
	$I_{OS}(+)$	20	---	---	+10 V	"	+10 V	"	---	"	"	"		11	I7	mA	$I_{OS}(+) = I7$	40.0		-40.0		mA
	$I_{OS}(-)$	21	---	---	"	"	-10 V	"	---	"	"	"		"	I8	"	$I_{OS}(-) = I8$		50.0		50.0	"
	I_{CC}	22	---	---	0 V	"	0 V	"	---	"	"	"		10	I9	"	$I_{CC} = I9$		6.5		6.5	"
	I_{EE}	23	---	---	"	"	"	"	---	"	"	"		9	I10	"	$I_{EE} = I10$	-6.5		-6.5		"
	CMRR(x)	24	---	---	-10 V	-10 V	+10 V	"	---	"	"	"		11	E11	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	70.0		60.0		dB
	CMRR(x)	25	---	---	0 V	0 V	"	"	---	"	"	"		"	E12	"						
	CMRR(x)	26	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E13	"						
	CMRR(y)	27	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E14	"	$CMRR(y) = 20 \log \frac{10}{\Delta E_y}$	"		"		"
	CMRR(y)	28	---	---	"	"	0 V	0 V	---	"	"	"		"	E15	"						
	CMRR(y)	29	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E16	"						
	CMRR(z)	30	---	---	"	"	0 V	---	-10 V	-10 V	"	"	K10, K12	"	E17	"	$CMRR(z) = 20 \log \frac{10}{\Delta E_z}$	"		"		"
	CMRR(z)	31	---	---	"	"	"	---	0 V	0 V	"	"	"	"	E18	"						
	CMRR(z)	32	---	---	"	"	"	---	+10 V	+10 V	"	"	"	"	E19	"						
	V_{OP}	33	---	---	+11 V	"	+11 V	0 V	---	0 V	"	"		"	E20	V	$V_{OP} = E20$	+11.0		+11.0		V
	V_{OP}	34	---	---	"	"	-11 V	"	---	"	"	"		"	E21	"	$V_{OP} = E21$		-11.0		-11.0	"
	PSRR1	35	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E22	"	$PSRR1 = \frac{(E22-E23)}{1.5}$	-10.0	10.0	10.0	10.0	$\frac{mV}{V}$
	PSRR1	36	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E23	"						
	PSRR2	37	---	---	"	"	"	"	---	"	"	"		"	E24	"	$PSRR2 = \frac{(E24-E25)}{3}$	-4.0	4.0	-4.0	4.0	"
	PSRR2	38	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E25	"						
	FT(x)	39	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E26	mV	$FT(x) = \left(\frac{E26+E27}{2}\right) - E5$	-35.0	35.0	-60.0	60.0	mV
	FT(x)	40	---	---	+10 V	"	"	"	---	"	"	"		"	E27	"						

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
1 T_A = +25°C	FT(y)	41	---	---	0 V	0 V	-10 V	0 V	---	0 V	-15 V	+15 V		11	E28	mV	$FT(y) = \left(\frac{E28+E29}{2}\right) - ES$	-12.0	12.0	-24.0	24.0	mV
	FT(y)	42	---	---	"	"	+10 V	"	---	"	"	"		"	E29	"						
	+NLx	43	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E4+E2)/2 -E29)/2)	-0.3	0.3	-0.5	0.5	%FS
	-NLx	44	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E3+E1)/2-E28)/2)	"	"	"	"	"
	+NLy	45	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E4+E3)/2-E27)/2)	-0.15	0.15	-0.2	0.2	"
	-NLy	46	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E1+E2)/2-E26)/2)	"	"	"	"	"
2 T_A = +125°C	MAx-y-	47	---	---	-10 V	0 V	-10 V	0 V	---	0 V	-15 V	+15 V		11	E30	V	$MAx\text{-}y\text{-} = \frac{(E30 \cdot 10)}{10} \times 100$	-1.0	1.0	-2.0	2.0	"
	MAx-y+	48	---	---	"	"	+10 V	"	---	"	"	"		"	E31	"	$MAx\text{-}y+ = \frac{(E31+10)}{10} \times 100$	"	"	"	"	"
	MAx+y-	49	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E32	"	$MAx+y- = \frac{(E32+10)}{10} \times 100$	"	"	"	"	"
	MAx+y+	50	---	---	"	"	+10 V	"	---	"	"	"		"	E33	"	$MAx+y+ = \frac{(E33-10)}{10} \times 100$	"	"	"	"	"
	$\frac{MAxy}{\Delta T}$	51	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{MAxy}{\Delta T} = \left[\frac{MAxy(125°C)-MAxy(25°C)}{100}\right]$	-0.01	0.01	-0.02	0.02	$\frac{\%FS}{\Delta T}$
	V_{IO}	52	---	---	0 V	"	0 V	"	---	"	"	"		11	E34	mV	$V_{IO} = E34$	-45.0	45.0	-80.0	80.0	mV
	$\frac{V_{IO}}{\Delta T}$	53	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO}}{\Delta T} = \frac{(E34-E5)}{100}$	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{°C}$
	$V_{IO(x)}$	54	---	---	"	"	-10 V	"	---	"	"	"		11	E35	mV	$V_{IO(x)} = \frac{(E35-E36)}{2}$	-25.0	25.0	-40.0	40.0	mV
	$V_{IO(x)}$	55	---	---	"	"	+10 V	"	---	"	"	"		"	E36	"						
	$\frac{V_{IO(x)}}{\Delta T}$	56	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(x)}}{\Delta T} = \frac{(V_{IO(x)}\ @125°C - V_{IO(x)}@25°C)}{100}$	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{°C}$
	$V_{IO(y)}$	57	---	---	-10 V	"	0 V	"	---	"	"	"		11	E37	mV	$V_{IO(y)} = \frac{(E37-E38)}{2}$	-25.0	25.0	-40.0	40.0	mV
	$V_{IO(y)}$	58	---	---	+10 V	"	"	"	---	"	"	"		"	E38	"						
	$\frac{V_{IO(y)}}{\Delta T}$	59	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(y)}}{\Delta T} = \frac{(V_{IO(y)}\ @125°C - V_{IO(y)}\ @25°C)}{100}$	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{°C}$

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
2 T_A = +125°C	$V_{IO}(z)$	60	---	---	+1 V	0 V	0 V	---	0 V	0 V	-15 V	+15 V	K10, K12	11	E39	mV	$V_{IO}(z) = \frac{E(39)}{10}$	-45.0	45.0	-80.0	80.0	mV
	$\frac{V_{IO}(z)}{\Delta T}$	61	---	---	"	"	"	---	"	"	"	"		---	---	---	$\frac{V_{IO}(z)}{\Delta T} = \frac{(V_{IO}(z)\ @125°C - V_{IO}(z)\ @25°C)}{100}$	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{°C}$
	$+I_{IB}(x)$	62	---	---	---	"	"	0 V	"	"	"	"	K5	14	I11	µA		-2.0	2.0	-2.0	2.0	µA
	$+I_{IB}(y)$	63	---	---	0 V	"	---	"	"	"	"	"	K6	"	I12	"		"	"	"	"	"
	$+I_{IB}(z)$	64	---	---	"	"	0 V	"	---	"	"	"	K9, K10, K12	15	I13	"		"	"	"	"	"
	$-I_{IB}(x)$	65	---	---	"	---	"	"	0 V	"	"	"	K2	14	I14	"		"	"	"	"	"
	$-I_{IB}(y)$	66	---	---	"	0 V	"	---	"	"	"	"	K4	"	I15	"		"	"	"	"	"
	$-I_{IB}(z)$	67	---	---	"	"	"	0 V	"	---	"	"	K8	15	I16	"		"	"	"	"	"
	$I_{IO}(x)$	68	---	---	"	"	"	"	"	0 V	"	"		---	---	---	$I_{IO}(x) = I14 - I11$	-250.0	250.0	-250.0	250.0	nA
	$I_{IO}(y)$	69	---	---	"	"	"	"	"	"	"	"		---	---	---	$I_{IO}(y) = I15 - I12$	"	"	"	"	"
	$I_{IO}(z)$	70	---	---	"	"	"	"	"	"	"	"		---	---	---	$I_{IO}(z) = I16 - I13$	"	"	"	"	"
	$I_{OS}(+)$	71	---	---	+10 V	"	+10 V	"	---	"	"	"		11	I17	mA	$I_{OS}(+) = I17$	-30.0		-30.0		mA
	$I_{OS}(-)$	72	---	---	"	"	-10 V	"	---	"	"	"		"	I18	"	$I_{OS}(-) = I18$		30.0		30.0	"
	I_{CC}	73	---	---	0 V	"	0 V	"	---	"	"	"		10	I19	"	$I_{CC} = I19$		6.5		6.5	"
	I_{EE}	74	---	---	"	"	"	"	---	"	"	"		9	I20	"	$I_{EE} = I20$	-6.5		-6.5		"
	CMRR(x)	75	---	---	-10 V	-10 V	+10 V	"	---	"	"	"		11	E40	mV	$CMRR(x) = 20 \log \left(\frac{10}{\Delta E_x} \right)$	70.0		60.0		dB
	CMRR(x)	76	---	---	0 V	0 V	"	"	---	"	"	"		"	E41	"						
	CMRR(x)	77	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E42	"						
	CMRR(y)	78	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E43	"	$CMRR(y) = 20 \log \left(\frac{10}{\Delta E_y} \right)$	"		"		"
	CMRR(y)	79	---	---	"	"	0 V	0 V	---	"	"	"		"	E44	"						
	CMRR(y)	80	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E45	"						

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
2 T_A = +125°C	CMRR(z)	81	---	---	+10 V	0 V	0 V	---	-10 V	-10 V	-15 V	+15 V	K10, K12	11	E46	mV	$CMRR(z) = 20 \log\left(\frac{10}{\delta E_z}\right)$	70.0		60.0		dB
	CMRR(z)	82	---	---	"	"	"	---	0 V	0 V	"	"	"	"	E47	"						
	CMRR(z)	83	---	---	"	"	"	---	+10 V	-10 V	"	"	"	"	E48	"						
	V_{OP}	84	---	---	+11 V	"	+11 V	0 V	---	0 V	"	"		"	E49	V	V_{OP} = E49	+11.0		+11.0		V
	V_{OP}	85	---	---	"	"	-11 V	"	---	"	"	"		"	E50	"	V_{OP} = E50		-11.0		-11.0	"
	PSRR1	86	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E51	"	$PSRR1 = \frac{(E51-E52)}{1.5}$	-10.0	10.0	-10.0	10.0	$\frac{mV}{V}$
	PSRR1	87	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E52	"						
	PSRR2	88	---	---	"	"	"	"	---	"	"	"		"	E53	"	$PSRR2 = \frac{(E54-E55)}{3}$	-4.0	4.0	-4.0	4.0	mV
	PSRR2	89	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E54	"						
	FT(x)	90	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E55	mV	$FT(x) = \left(\frac{E56+E57}{2}\right) - E34$	-35.0	35.0	-60.0	60.0	"
	FT(x)	91	---	---	+10 V	"	"	"	---	"	"	"		"	E56	"						
	FT(y)	92	---	---	0 V	"	-10 V	"	---	"	"	"		"	E57	"	$FT(y) = \left(\frac{E57+E58}{2}\right) - E34$	-12.0	12.0	-24.0	24.0	"
	FT(y)	93	---	---	"	"	+10 V	"	---	"	"	"		"	E58	"						
	+NLx	94	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E33+E31)/2-E58)/2)	-0.3	0.3	-0.5	0.5	%FS
	-NLx	95	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E32+E30)/2-E57)/2)	"	"	"	"	"
	+NLy	96	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E33+E32)/2-E56/2)	-0.15	0.15	-0.2	0.2	"
	-NLy	97	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E30+E31)/2-E55)/2)	"	"	"	"	"
3 T_A = -55°C	MAx-y-	98	---	---	-10 V	0 V	-10 V	0 V	---	0 V	-15 V	+15 V		11	E59	V	$MAx\text{-}y\text{-} = \frac{(E59-10)}{10} \times 100$	-1.0	1.0	-2.0	2.0	"
	MAx-y+	99	---	---	"	"	+10 V	"	---	"	"	"		"	E60	"	$MAx\text{-}y+ = \frac{(E60+10)}{10} \times 100$	"	"	"	"	"
	MAx+y-	100	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E61	"	$MAx+y- = \frac{(E61+10)}{10} \times 100$	"	"	"	"	"
	MAx+y+	101	---	---	"	"	+10 V	"	---	"	"	"		"	E62	"	$MAx+y+ = \frac{(E62-10)}{10} \times 100$	"	"	"	"	"

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
3 T_A = -55°C	$\frac{MA\ xy}{\Delta T}$	102	---	---	+10 V	0 V	+10 V	0 V	---	0 V	-15 V	+15 V		---	---	---	$\frac{MA\ xy}{\Delta T} = \left[\frac{(MAxy(-55°C)-MAxy(25°C))}{80}\right]$	-0.01	0.01	-0.02	0.02	%FS/ΔT
	V_{OIO}	103	---	---	0 V	"	0 V	"	---	"	"	"		11	E63	mV	V_{OIO} = E63	-39.0	39.0	-70.0	70.0	mV
	$\frac{V_{OIO}}{\Delta T}$	104	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{OIO}}{\Delta T} = \frac{(E63-E5)}{80}$	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{°C}$
	$V_{IO(x)}$	105	---	---	"	"	-10 V	"	---	"	"	"		11	E64	mV	$V_{IO(x)} = \frac{(E64-E65)}{2}$	-22.0	22.0	-36.0	36.0	mV
	$V_{IO(x)}$	106	---	---	"	"	+10 V	"	---	"	"	"		"	E65	"						
	$\frac{V_{IO(x)}}{\Delta T}$	107	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(x)}}{\Delta T} = \frac{(V_{IO(x)}\ @-55°C - V_{IO(x)}@25°C)}{80}$	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{°C}$
	$V_{IO(y)}$	108	---	---	-10 V	"	0 V	"	---	"	"	"		11	E66	mV	$V_{IO(y)} = \frac{(E66-E67)}{2}$	-22.0	22.0	-36.0	36.0	mV
	$V_{IO(y)}$	109	---	---	+10 V	"	"	"	---	"	"	"		"	E67	"						
	$\frac{V_{IO(y)}}{\Delta T}$	110	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(y)}}{\Delta T} = \frac{(V_{IO(y)}\ @-55°C - V_{IO(y)}\ @25°C)}{80}$	-150.0	150.0	-200.0	200.0	$\frac{\mu V}{°C}$
	$V_{IO(z)}$	111	---	---	+ 1 V	"	"	---	0 V	"	"	"	K10, K12	11	E68	mV	$V_{IO(z)} = \frac{E(68)}{10}$	-39.0	39.0	-70.0	70.0	mV
	$\frac{V_{IO(z)}}{\Delta T}$	112	---	---	"	"	"	---	"	"	"	"		---	---	---	$\frac{V_{IO(z)}}{\Delta T} = \frac{(V_{IO(z)}\ @-55°C - V_{IO(z)}\ @25°C)}{80}$	-300.0	300.0	-500.0	500.0	$\frac{\mu V}{°C}$
	$+I_{IB(x)}$	113	---	---	---	"	"	0 V	"	"	"	"	K5	14	I21	μA		-2.5	2.5	-2.5	2.5	μA
	$+I_{IB(y)}$	114	---	---	0 V	"	---	"	"	"	"	"	K6	"	I22	"		"	"	"	"	"
	$+I_{IB(z)}$	115	---	---	"	"	0 V	"	---	"	"	"	K9, K10, K12	15	I23	"		"	"	"	"	"
	$-I_{IB(x)}$	116	---	---	"	---	"	"	0 V	"	"	"	K2	14	I24	"		"	"	"	"	"
	$-I_{IB(y)}$	117	---	---	"	0 V	"	---	"	"	"	"	K4	"	I25	"		"	"	"	"	"
	$-I_{IB(z)}$	118	---	---	"	"	"	0 V	"	---	"	"	K8	15	I26	"		"	"	"	"	"
	$I_{IO(x)}$	119	---	---	"	"	"	"	"	0 V	"	"		---	---	---	$I_{IO(x)} = I24-I21$	-300.0	300.0	-300.0	300.0	nA
	$I_{IO(y)}$	120	---	---	"	"	"	"	"	"	"	"		---	---	---	$I_{IO(y)} = I25-I22$	"	"	"	"	"
	$I_{IO(z)}$	121	---	---	"	"	"	"	"	"	"	"		---	---	---	$I_{IO(z)} = I26-I23$	"	"	"	"	"

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	Unit
3 T_A = -55°C	$I_{OS}(+)$	122	---	---	+10 V	0 V	+10 V	0 V	---	0 V	-15 V	+15 V		11	127	mA	$I_{OS}(+) = I27$	-40.0		-40.0		mA
	$I_{OS}(-)$	123	---	---	"	"	-10 V	"	---	"	"	"		"	128	"	$I_{OS}(-) = I28$		50.0		50.0	"
	I_{CC}	124	---	---	0 V	"	0 V	"	---	"	"	"		10	129	"	$I_{CC} = I29$		6.5		6.5	"
	I_{EE}	125	---	---	"	"	"	"	---	"	"	"		9	130	"	$I_{EE} = I30$	-6.5		-6.5		"
	CMRR(x)	126	---	---	-10 V	-10 V	+10 V	"	---	"	"	"		11	E69	mV	$CMRR(x) = 20 \log \left(\frac{10}{\Delta E_x}\right)$	65.0		54.0		dB
	CMRR(x)	127	---	---	0 V	0 V	"	"	---	"	"	"		"	E70	"						
	CMRR(x)	128	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E71	"						
	CMRR(y)	129	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E72	"	$CMRR(y) = 20 \log \left(\frac{10}{\Delta E_y}\right)$	"		"		"
	CMRR(y)	130	---	---	"	"	0 V	0 V	---	"	"	"		"	E73	"						
	CMRR(y)	131	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E74	"						
	CMRR(z)	132	---	---	"	"	0 V	---	-10 V	-10 V	"	"	K10, K12	"	E75	"	$CMRR(z) = 20 \log \left(\frac{10}{\Delta E_z}\right)$	"		"		"
	CMRR(z)	133	---	---	"	"	"	---	0 V	0 V	"	"	"	"	E76	"						
	CMRR(z)	134	---	---	"	"	"	---	+10 V	+10 V	"	"	"	"	E77	"						
	V_{OP}	135	---	---	+11 V	"	+11 V	0 V	---	0 V	"	"		"	E78	V	$V_{OP} = E78$	+11.0		+11.0		V
	V_{OP}	136	---	---	"	"	-11 V	"	---	"	"	"		"	E79	"	$V_{OP} = E79$		-11.0		11.0	"
	PSRR1	137	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E80	"	$PSRR1 = \frac{(E80-E81)}{1.5}$	-10.0	10.0	-10.0	10.0	mV/V
	PSRR1	138	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E81	"						
	PSRR2	139	---	---	"	"	"	"	---	"	"	"		"	E82	"	$PSRR2 = \frac{(E82-E83)}{1.5}$	-4.0	4.0	-4.0	4.0	mV
	PSRR2	140	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E83	"						
	FT(x)	141	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E84	mV	$FT(x) = \left(\frac{E84+E85}{2}\right) - E63$	-35.0	35.0	-60.0	60.0	"
	FT(x)	142	---	---	+10 V	"	"	"	---	"	"	"		"	E85	"						
	FT(y)	143	---	---	0 V	"	-10 V	"	---	"	"	"		"	E86	"	$FT(y) = \left(\frac{E86+E87}{2}\right) - E63$	-12.0	12.0	-24.0	24.0	"
	FT(y)	144	---	---	"	"	+10 V	"	---	"	"	"		"	E87	"						

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
3 T_A = -55°C	+NLx	145	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E62+E60)/2-E87)/2)	0.3	0.3	-0.5	0.5	IFS
	-NLx [2]	146	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E61+E59)/2-E86)/2)	"	"	"	"	"
	+NLy	147	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E62+E61)/2-E85/2)	-0.15	0.15	-0.2	0.2	"
	-NLy	148	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E59+E60)/2-E84)/2)	"	"	"	"	"
4 T_A = +25°C	SR(+)	149	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	18.0		18.0		$\frac{V}{\mu s}$
	SR(+)	150	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs						
	SR(-)	151	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-)= ΔVo(-)/Δt(-)	"		"		"
	SR(-)	152	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs						
	AE(x)	153	2 Vp-p	---	---	"	"	"	---	"	"	"	K1, K13, K14	"	f1	kHz	AE(x) = f1 θ(-1%) Eo	75.0		75.0		kHz
	AE(y)	154	"	---	+10 V	"	---	"	---	"	"	"	K3, K13, K14	"	f2	"	AE(y) = f2 θ(-1%) Eo	"		"		"
5 T_A = +125°C	SR(+)	155	---	20 Vp-p f = 10kHz	---	0 V	+10 V	0 V	---	0 V	-15 V	+15 V	K1, K14, K15	11	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	18.0		18.0		$\frac{V}{\mu s}$
	SR(+)	156	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs						
	SR(-)	157	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	"		"		"
	SR(-)	158	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs						

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits				Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Device 01 Min	Device 01 Max	Device 02 Min	Device 02 Max	
6 T_A = -55°C	SR(+)	159	---	20 Vp-p f = 10kHz	---	0 V	+10 V	0 V	---	0 V	-15 V	+15 V	K1, K14, K15	11	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	18.0		18.0		$\frac{V}{\mu s}$
	SR(+)	160	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	µs						
	SR(-)	161	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	"		"		"
	SR(-)	162	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	µs						
7 T_A = +25°C	HI(B3)	163	---	---	--	---	---	---	---	---	---	---		---	---	---	See figure 6 BW = 10 Hz to 100 kHz		1.5		1.5	mV rms
9 T_A = +25°C	ts(+)	164	---	20 Vp-p f = 10KHz	---	0 V	+10 V	0 V	---	0 V	-15 V	+15 V	K1, K4, K15	11	ts(+)	µs	See figure 5, waveform 1		3.0		3.0	µs
	ts(-)	165	---	"	---	"	"	"	---	"	"	"	"	"	ts(-)	"	See figure 5, waveform 2		"		"	"

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
1 T_A = +25°C	MAx-y-	1	---	---	-10 V	0 V	-10 V	0 V	---	GND	-15 V	+15 V		11	E1	V	$MAx\text{-}y\text{-} = \frac{(E1-10)}{10} \times 100$	1.0	1.0	%
	MAx-y+	2	---	---	"	"	+10 V	"	---	"	"	"		"	E2	"	$MAx\text{-}y+ = \frac{(E2+10)}{10} \times 100$	"	"	"
	MAx+y-	3	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E3	"	$MAx+y- = \frac{(E3+10)}{10} \times 100$	"	"	"
	MAx+y+	4	---	---	"	"	+10 V	"	---	"	"	"		"	E4	"	$MAx+y+ = \frac{(E4-10)}{10} \times 100$	"	"	"
	V_{OIO}	5	---	---	0 V	"	"	"	---	"	"	"		"	E5	mV	$V_{OIO} = E5$	-30.0	30.0	mV
	$V_{IO}(x)$	6	---	---	"	"	"	"	---	"	"	"		"	E6	"	$V_{IO}(x) = \frac{(E6-E7)}{2}$	-50.0	50.0	"
	$V_{IO}(x)$	7	---	---	"	"	-10 V	"	---	"	"	"		"	E7	"				
	$V_{IO}(y)$	8	---	---	+10 V	"	0 V	"	---	"	"	"		"	E8	"	$V_{IO}(y) = \frac{(E8-E9)}{2}$	"	"	"
	$V_{IO}(y)$	9	---	---	-10 V	"	"	"	---	"	"	"		"	E9	"				
	$I_{IB}(z)$	10	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	I1	µA	$I_{IB}(z) = I1$	-12.0	12.0	µA
	$+I_{IB}(x)$	11	---	---	---	"	0 V	"	---	"	"	"	K5	8	I2	"		-2.0	2.0	"
	$+I_{IB}(y)$	12	---	---	0 V	"	---	"	---	"	"	"	K6	"	I3	"		"	"	"
	$-I_{IB}(x)$	13	---	---	"	---	0 V	"	---	"	"	"	K2	"	I4	"		"	"	"
	$-I_{IB}(y)$	14	---	---	"	0 V	"	---	---	"	"	"	K4	"	I5	"		"	"	"
	$I_{IO}(x)$	15	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO}(x) = I2-I4$	-200.0	200.0	nA
	$I_{IO}(y)$	16	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO}(y) = I3-I5$	"	"	"

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
1 T_A = +25°C	$I_{OS}(+)$	17	---	---	+10 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		11	I6	mA	$I_{OS}(+) = I6$	-45.0		mA
	$I_{OS}(-)$	18	---	---	"	"	-10 V	"	---	"	"	"		"	I7	"	$I_{OS}(-) = I7$		80.0	"
	I_{CC}	19	---	---	0 V	"	0 V	"	---	"	"	"		10	I8	"	$I_{CC} = I8$		6.5	"
	I_{EE}	20	---	---	"	"	"	"	---	"	"	"		9	I9	"	$I_{EE} = I9$	-6.5		"
	CMRR(x)	21	---	---	-10 V	-10 V	+10 V	"	---	"	"	"		11	E10	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	90.0		dB
	CMRR(x)	22	---	---	0 V	0 V	"	"	---	"	"	"		"	E11	"				
	CMRR(x)	23	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E12	"				
	CMRR(y)	24	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E13	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	25	---	---	"	"	0 V	0 V	---	"	"	"		"	E14	"				
	CMRR(y)	26	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E15	"				
	ERR1	27	---	---	-10 V	"	-5 V	0 V	---	"	"	"		"	E16	V		4.9	5.1	V
	ERR2	28	---	---	+10 V	"	"	"	---	"	"	"		"	E17	"		-5.1	-4.9	"
	ERR3	29	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E18	"		"	"	"
	ERR4	30	---	---	+10 V	"	"	"	---	"	"	"		"	E19	"		4.9	5.1	"
	ERR5	31	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E20	"		"	"	"
	ERR6	32	---	---	"	"	+10 V	"	---	"	"	"		"	E21	"		-5.1	-4.9	"
	ERR7	33	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E22	"		"	"	"
	ERR8	34	---	---	"	"	+10 V	"	---	"	"	"		"	E23	"		4.9	5.1	"
	V_{OP}	35	---	---	+11 V	"	+11 V	"	---	"	"	"		"	E24	"	$V_{OP} = E24$	+10.0		"
	V_{OP}	36	---	---	"	"	-11 V	"	---	"	"	"		"	E25	"	$V_{OP} = E25$		-10.0	"
	PSRR1	37	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E26	"	$PSRR1 = \frac{(E26-E27)}{1.5}$	-50.0	50.0	$\frac{mV}{V}$
	PSRR1	38	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E27	"				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
1 T_A = +25°C	PSRR2	39	---	---	+5 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		11	E28	V	$PSRR2 = \frac{(E28-E29)}{3.0}$	50.0	50.0	$\frac{mV}{V}$
	PSRR2	40	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E29	"				
	FT(x)	41	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E30	mV	$FT(x) = \left(\frac{E30+E31}{2}\right) - E5$	100.0	100.0	mV
	FT(x)	42	---	---	+10 V	"	"	"	---	"	"	"		"	E31	"				
	FT(y)	43	---	---	0 V	"	-10 V	"	---	"	"	"		"	E32	"	$FT(y) = \left(\frac{E32+E33}{2}\right) - E5$	-80.0	80.0	"
	FT(y)	44	---	---	"	"	+10 V	"	---	"	"	"		"	E33	"				
	+NLx(A)	45	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E21-(((E2-E4)/4+(((E2+E4)/2)+E33)/2	-0.6	0.6	%FS
	+NLx(B)	46	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E23+(((E2-E4)/4+(((E2+E4)/2)+E33)/2	"	"	"
	-NLx(A)	47	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E20-(((E1-E3)/4+(((E1+E3)/2+E32)/2	"	"	"
	-NLx(B)	48	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E22+(((E1-E3)/4+(((E1+E3)/2+E32)/2	"	"	"
	+NLy(A)	49	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E17-(((E3-E4)/4+(((E4+E3)/2+E31)/2	-0.4	0.4	"
	+NLy(B)	50	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E19+(((E3-E4)/4+(((E4+E3)/2+E31)/2	"	"	"
	-NLy(A)	51	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E18-(((E2-E1)/4+(((E2+E1)/2+E30)/2	"	"	"
	-NLy(B)	52	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E16+(((E2-E1)/4+(((E2+E1)/2+E30)/2	"	"	"
	+NLx	53	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E4+E2)/2)-E33)/2	-0.6	0.6	"
	-NLx	54	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E3+E1)/2)-E32)/2	"	"	"
	+NLy	55	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E4+E3)/2)-E31)/2	-0.4	0.4	"
	-NLy	56	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E1+E2)/2)-E30)/2	"	"	"
2 T_A = +125°C	MAx-y-	57	---	---	-10 V	0 V	-10 V	0 V	---	GND	-15 V	+15 V		11	E34	V	$MAx\text{-}y\text{-} = \frac{(E34-10)}{10} \times 100$	-4.0	4.0	"
	MAx-y+	58	---	---	"	"	+10 V	"	---	"	"	"		"	E35	"	$MAx\text{-}y+ = \frac{(E35+10)}{10} \times 100$	"	"	"
	MAx+y-	59	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E36	"	$MAx+y- = \frac{(E36+10)}{10} \times 100$	"	"	"
	MAx+y+	60	---	---	"	"	+10 V	"	---	"	"	"		"	E37	"	$MAx+y+ = \frac{(E37-10)}{10} \times 100$	"	"	"

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
2 T_A = +125°C	$\frac{MAxy}{\Delta T}$	61	---	---	+10 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		---	---	---	$\frac{MAxy}{\Delta T} = \frac{(MAxy\ (125°C) - MAxy\ (25°C)}{100}$	-0.04	0.04	%FS
	V_{OIO}	62	---	---	0 V	"	0 V	"	---	"	"	"		11	E38	mV	$V_{OIO} = E38$	-230.0	230.0	mV
	$\frac{V_{OIO}}{\Delta T}$	63	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{OIO}}{\Delta T} = \frac{(E38-E5)}{100}$	-2.0	2.0	$\frac{\mu V}{°C}$
	$V_{IO(x)}$	64	---	---	"	"	-10 V	"	---	"	"	"		11	E39	mV	$V_{IO(x)} = \frac{(E39-E40)}{2}$	-100.0	100.0	mV
	$V_{IO(x)}$	65	---	---	"	"	+10 V	"	---	"	"	"		"	E40	"				
	$\frac{V_{IO(x)}}{\Delta T}$	66	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(x)}}{\Delta T} = \frac{(V_{IO(x)}\ @125°C - V_{IO(x)}\ @25°C)}{100}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$V_{IO(y)}$	67	---	---	-10 V	"	0 V	"	---	"	"	"		11	E41	mV	$V_{IO(y)} = \frac{(E41-E42)}{2}$	-100.0	100.0	mV
	$V_{IO(y)}$	68	---	---	+10 V	"	"	"	---	"	"	"		"	E42	"				
	$\frac{V_{IO(y)}}{\Delta T}$	69	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(y)}}{\Delta T} = \frac{(V_{IO(y)}\ @125°C - V_{IO(y)}\ @25°C)}{100}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$I_{IB(z)}$	70	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	I10	µA	$I_{IB(z)} = I10$	-12.0	12.0	µA
	$+I_{IB(x)}$	71	---	---	---	"	0 V	"	---	"	"	"	K5	8	I11	"		-2.0	2.0	"
	$+I_{IB(y)}$	72	---	---	0 V	"	---	"	---	"	"	"	K6	"	I12	"		"	"	"
	$-I_{IB(x)}$	73	---	---	"	---	0 V	"	---	"	"	"	K2	"	I13	"		"	"	"
	$-I_{IB(y)}$	74	---	---	"	0 V	"	"	---	"	"	"	K4	"	I14	"		"	"	"
	$I_{IO(x)}$	75	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO(x)} = I11-I13$	-200.0	200.0	nA
	$I_{IO(y)}$	76	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO(y)} = I12-I14$	"	"	"
	$I_{OS(+)}$	77	---	---	+10 V	"	+10 V	"	---	"	"	"		11	I15	mA	$I_{OS(+)} = I15$	-30.0		mA
	$I_{OS(-)}$	78	---	---	"	"	-10 V	"	---	"	"	"		"	I16	"	$I_{OS(-)} = I16$		30.0	"
	I_{CC}	79	---	---	0 V	"	0 V	"	---	"	"	"		10	I17	"	$I_{CC} = I17$		6.5	"
	I_{EE}	80	---	---	"	"	"	"	---	"	"	"		9	I18	"	$I_{EE} = I18$	-6.5		"

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
4 T_A = +125°C	CMRR(x)	81	---	---	-10 V	-10 V	+10 V	0 V	---	GND	-15 V	+15 V		11	E43	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	50.0		dB
	CMRR(x)	82	---	---	0 V	0 V	"	"	---	"	"	"		"	E44	"				
	CMRR(x)	83	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E45	"				
	CMRR(y)	84	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E46	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	85	---	---	"	"	0 V	0 V	---	"	"	"		"	E47	"				
	CMRR(y)	86	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E48	"				
	ERR1	87	---	---	-10 V	"	5 V	0 V	---	"	"	"		"	E49	V		4.6	5.4	V
	ERR2	88	---	---	+10 V	"	"	"	---	"	"	"		"	E50	"		5.4	4.6	"
	ERR3	89	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E51	"		"	"	"
	ERR4	90	---	---	+10 V	"	"	"	---	"	"	"		"	E52	"		4.6	5.4	"
	ERR5	91	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E53	"		"	"	"
	ERR6	92	---	---	"	"	+10 V	"	---	"	"	"		"	E54	"		5.4	4.6	"
	ERR7	93	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E55	"		"	"	"
	ERR8	94	---	---	"	"	+10 V	"	---	"	"	"		"	E56	"		4.6	5.4	"
	V_{OP}	95	---	---	+11 V	"	+11 V	"	---	"	"	"		"	E57	"	V_{OP} = E57	+10.0		"
	V_{OP}	96	---	---	"	"	-11 V	"	---	"	"	"		"	E58	"	V_{OP} = E58		10.0	"
	PSRR1	97	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E59	"	$PSRR1 = \frac{(E59-E60)}{1.5}$	-50.0	50.0	$\frac{mV}{V}$
	PSRR1	98	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E60	"				
	PSRR2	99	---	---	"	"	"	"	---	"	"	"		"	E61	"	$PSRR2 = \frac{(E61-E62)}{3.0}$	"	"	"
	PSRR2	100	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E62	"				
	FT(x)	101	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E63	mV	$FT(x) = \left(\frac{E63+E64}{2}\right) - E38$	-100.0	100.0	mV
	FT(x)	102	---	---	+10 V	"	"	"	---	"	"	"		"	E64	"				
	FT(y)	103	---	---	0 V	"	-10 V	"	---	"	"	"		"	E65	"	$FT(y) = \left(\frac{E65+E66}{2}\right) - E38$	-80.0	80.0	"
	FT(y)	104	---	---	"	"	+10 V	"	---	"	"	"		"	E66	"				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
2 T_A = +125°C	+NLx(A)	105	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E54-((E35-E37)/4*(((E35*E37)/2)*E66)/2	-0.6	0.6	%FS
	+NLx(B)	106	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E56+((E35-E37)/4*(((E35*E37)/2)*E66)/2	"	"	"
	-NLx(A)	107	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E53-((E34-E36)/4*(((E34*E36)/2)*E65)/2	"	"	"
	-NLx(B)	108	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E55+((E34-E36)/4*(((E34*E36)/2)*E65)/2	"	"	"
	+NLy(A)	109	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E50-((E36-E37)/4*(((E36*E37)/2)*E64)/2	-0.4	0.4	"
	+NLy(B)	110	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E52+((E36-E37)/4*(((E36*E37)/2)*E64)/2	"	"	"
	-NLy(A)	111	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E51-((E35-E34)/4*(((E35*E34)/2)*E63)/2	"	"	"
	-NLy(B)	112	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E49+((E35-E34)/4*(((E35*E34)/2)*E63)/2	"	"	"
	+NLx	113	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E37*E35)/2)-E66)/2	-0.6	0.6	"
	-NLx	114	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E36*E34)/2-E65)/2	"	"	"
	+NLy	115	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E37*E36)/2)-E64)/2	-0.4	0.4	"
	-NLy	116	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E34*E35)/2)-E63)/2	"	"	"
3 T_A = -55°C	MAx-y-	117	---	---	-10 V	0 V	-10 V	0 V	---	GND	-15 V	+15 V		11	E67	V	$MAx\text{-}y\text{-} = \frac{(E67-10)}{10} \times 100$	-4.0	4.0	"
	MAx-y+	118	---	---	"	"	+10 V	"	---	"	"	"		"	E68	"	$MAx\text{-}y+ = \frac{(E68*10)}{10} \times 100$	"	"	"
	MAx+y-	119	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E69	"	$MAx+y- = \frac{(E69*10)}{10} \times 100$	"	"	"
	MAx+y+	120	---	---	"	"	+10 V	"	---	"	"	"		"	E70	"	$MAx+y+ = \frac{(E70-10)}{10} \times 100$	"	"	"
	$\frac{MAxy}{\Delta T}$	121	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{MAxy}{\Delta T} = \left[\frac{(MAxy\ (-55°C) - MAxy\ (25°C)}{80}\right]$	-0.04	0.04	$\frac{\%FS}{\Delta T}$
	V_{OIO}	122	---	---	0 V	"	0 V	"	---	"	"	"		11	E71	mV	V_{OIO} = E71	-190.0	190.0	mV
	$\frac{V_{OIO}}{\Delta T}$	123	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{OIO}}{\Delta T} = \frac{(E71-E5)}{80}$	-2.0	2.0	$\frac{mV}{°C}$
	$V_{IO}(x)$	124	---	---	"	"	-10 V	"	---	"	"	"		11	E72	mV	$V_{IO}(x) = \frac{(E72-E75)}{2}$	-100.0	100.0	mV
	$V_{IO}(x)$	125	---	---	"	"	+10 V	"	---	"	"	"		"	E73	"				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
3 T_A = -55°C	$\frac{V_{IO}(x)}{\Delta T}$	126	---	---	0 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		---	---	---	$\frac{V_{IO}(x)}{\Delta T} = \frac{(V_{IO}(x)\ @\ -55°C - V_{IO}(x)\ @\ 25°C)}{\Delta}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$V_{IO}(y)$	127	---	---	-10 V	"	0 V	"	---	"	"	"		11	E74	mV	$V_{IO}(y) = \frac{(E74-E75)}{2}$	-100.0	100.0	mV
	$V_{IO}(y)$	128	---	---	+10 V	"	"	"	---	"	"	"		"	E75	"				
	$\frac{V_{IO}(y)}{\Delta T}$	129	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO}(y)}{\Delta T} = \frac{(V_{IO}(y)\ @\ -55°C - V_{IO}(y)\ @\ 25°C)}{80}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$I_{IB}(z)$	130	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	119	μA	$I_{IB}(z) = 119$	-12.0	12.0	μA
	$+I_{IB}(x)$	131	---	---	---	"	0 V	"	---	"	"	"	K5	8	120	"		-4.0	4.0	"
	$+I_{IB}(y)$	132	---	---	0 V	"	---	"	---	"	"	"	K6	"	121	"		"	"	"
	$-I_{IB}(x)$	133	---	---	"	---	0 V	"	---	"	"	"	K7	"	122	"		"	"	"
	$-I_{IB}(y)$	134	---	---	"	0 V	"	---	---	"	"	"	K4	"	123	"		"	"	"
	$I_{IO}(x)$	135	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO}(x) = 120-122$	-400.0	400.0	nA
	$I_{IO}(y)$	136	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO}(y) = 121-123$	"	"	"
	$I_{OS}(+)$	137	---	---	+10 V	"	+10 V	"	---	"	"	"		11	124	mA	$I_{OS}(+) = 124$	-45.0		mA
	$I_{OS}(-)$	138	---	---	"	"	-10 V	"	---	"	"	"		"	125	"	$I_{OS}(-) = 125$		30.0	"
	I_{CC}	139	---	---	0 V	"	0 V	"	---	"	"	"		10	126	"	$I_{CC} = 126$		6.5	"
	I_{EE}	140	---	---	"	"	"	"	---	"	"	"		9	127	"	$I_{EE} = 127$	-6.5		"
	CMRR(x)	141	---	---	-10 V	-10 V	+10 V	"	---	"	"	"		11	E76	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	80		dB
	CMRR(x)	142	---	---	0 V	0 V	"	"	---	"	"	"		"	E77	"				
	CMRR(x)	143	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E78	"				
	CMRR(y)	144	---	---	"	0 V	-10 V	-10 V	---	"	"	"		"	E79	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	145	---	---	"	"	0 V	0 V	---	"	"	"		"	E80	"				
	CMRR(y)	146	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E81	"				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
3 T_A = -55°C	ERR1	147	---	---	-10 V	0 V	-5 V	0 V	---	GND	-15 V	+15 V		11	E82	V		4.6	5.4	V
	ERR2	148	---	---	+10 V	"	"	"	---	"	"	"		"	E83	"		-5.4	4.6	"
	ERR3	149	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E84	"		"	"	"
	ERR4	150	---	---	+10 V	"	"	"	---	"	"	"		"	E85	"		4.6	5.4	"
	ERR5	151	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E86	"		"	"	"
	ERR6	152	---	---	"	"	+10 V	"	---	"	"	"		"	E87	"		-5.4	-4.6	"
	ERR7	153	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E88	"		"	"	"
	ERR8	154	---	---	"	"	+10 V	"	---	"	"	"		"	E89	"		4.6	5.4	"
	V_{OP}	155	---	---	+11 V	"	+11 V	"	---	"	"	"		"	E90	"	V_{OP} = E90	+10.0		"
	V_{OP}	156	---	---	"	"	-11 V	"	---	"	"	"		"	E91	"	V_{OP} = E91		-10.0	"
	PSRR1	157	---	---	"	"	"	"	---	"	-13.5 V	+13.5 V		"	E92	"	$PSRR1 = \frac{(E92-E93)}{1.5}$	-50.0	50.0	mV/V
	PSRR1	158	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E93	"				
	PSRR2	159	---	---	"	"	"	"	---	"	"	"		"	E94	"	$PSRR2 = \frac{(E94-E95)}{3.0}$	"	"	"
	PSRR2	160	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E95	"				
	FT(x)	161	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E96	mV	$FT(x) = \left(\frac{E96+E97}{2}\right) - E71$	-100.0	100.0	mV
	FT(x)	162	---	---	+10 V	"	"	"	---	"	"	"		"	E97	"				
	FT(y)	163	---	---	0 V	"	-10 V	"	---	"	"	"		"	E98	"	$FT(y) = \left(\frac{E98+E99}{2}\right) - E71$	-80.0	80.0	"
	FT(y)	164	---	---	"	"	+10 V	"	---	"	"	"		"	E99	"				
	+NLx(A)	165	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E87-((E68-E70)/4+(((E68+E70)/2)+E99)/2	-0.6	0.6	%FS
	+NLx(B)	166	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E89+((E68-E70)/4+(((E68+E70)/2)+E99)/2	"	"	"
	-NLx(A)	167	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E86-((E67-E69)/4+(((E67+E69)/2)+E98)/2	"	"	"
	-NLx(B)	168	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E88+((E67-E69)/4+(((E67+E69)/2)+E98)/2	"	"	"
	+NLy(A)	169	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E83-((E69-E70)/4+(((E69+E70)/2)+E97)/2	-0.4	0.4	"
	+NLy(B)	170	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E85+((E69-E70)/4+(((E69+E70)/2)+E97)/2	"	"	"

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
3 T_A = -55°C	-NLy(A)	171	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E84-(((E68-E67)/4*(((E68*E67)/2)*E96)/2	-0.4	0.4	V
	-NLy(B)	172	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E82*(((E68-E67)/4*(((E68*E67)/2)*E96)/2	"	"	"
	*NLx	173	---	---	---	---	---	---	---	---	---	---		---	---	---	*NLx = (((E70*E68)/2)-E99)/2	-0.6	0.6	"
	-NLx	174	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E69*E67)/2)-E98)/2	"	"	"
	*NLy	175	---	---	---	---	---	---	---	---	---	---		---	---	---	*NLy = (((E70*E69)/2)-E97)/2	-0.4	0.4	"
	-NLy	176	---	---	---	---	---	---	---	---	---	---					-NLy = (((E67*E68)/2)-E96)/2	"	"	"
4 T_A = +25°C	SR(+)	177	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	25.0		V/μs
	SR(+)	178	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs				
	SR(-)	179	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	35.0		"
	SR(-)	180	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs				
	AE(x)	181	2 Vp-p	---	---	"	"	"	---	"	"	"	K1, K12, K13, K14	"	f1	kHz	AE(x) = f1 θ(-1%) Eo	70.0		kHz
	AE(y)	182	"	---	+10 V	"	---	"	---	"	"	"	"	"	f2	"	AE(y) = f2 θ(-1%) Eo	"		"
5 T_A = +125°C	SR(+)	183	---	20 Vp-p f = 10kHz	---	"	+10 V	"	---	"	"	"	K1, K12, K15	"	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	25.0		V/μs
	SR(+)	184	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs				
	SR(-)	185	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	15.0		"
	SR(-)	186	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
5 T_C = -55°C	SR(+)	187	---	20 Vp-p f = 10KHz	---	0 V	+10 V	0 V	---	GND	-15 V	+15 V	K1, K12, K15	11	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	25.0		V/μs
	SR(+)	188	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs				
	SR(-)	189	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	35.0		"
	SR(-)	190	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs				
7 T_A = +25°C	NI(BB)	191	---	---	--	---	---	---	---	---	---	---		---	---	---	See figure 6 BW = 10 Hz to 100 kHz		3.0	mVrms
9 T_A = +25°C	ts(+)	192	---	20 Vp-p f = 10KHz	---	0 V	+10 V	0 V	---	GND	-15 V	+15 V	K1, K12, K15	11	ts(+)	μs	See figure 5, waveform 1		2.0	μs
	ts(-)	193	---	"	---	"	"	"	---	"	"	"	"	"	ts(-)	"	See figure 5, waveform 2		"	"

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TABLE III. Group A inspection for device type 04.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
1 T_A = +25°C	$MAx\text{-}y\text{-}$	1	---	---	-10 V	0 V	-10 V	0 V	---	GND	-15 V	+15 V		11	E1	V	$MAx\text{-}y\text{-} = \frac{(E1-10)}{10} \times 100$	-1.0	1.0	FS
	$MAx\text{-}y\text{+}$	2	---	---	"	"	+10 V	"	---	"	"	"		"	E2	"	$MAx\text{-}y\text{+} = \frac{(E2+10)}{10} \times 100$	"	"	"
	$MAx\text{+}y\text{-}$	3	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E3	"	$MAx\text{+}y\text{-} = \frac{(E3+10)}{10} \times 100$	"	"	"
	$MAx\text{+}y\text{+}$	4	---	---	"	"	+10 V	"	---	"	"	"		"	E4	"	$MAx\text{+}y\text{+} = \frac{(E4-10)}{10} \times 100$	"	"	"
	V_{OIO}	5	---	---	0 V	"	"	"	---	"	"	"		"	E5	mV	$V_{OIO} = E5$	-30.0	30.0	mV
	$V_{IO}(x)$	6	---	---	"	"	"	"	---	"	"	"		"	E6	"	$V_{IO}(x) = \frac{(E6-E7)}{2}$	-40.0	40.0	"
	$V_{IO}(x)$	7	---	---	"	"	-10 V	"	---	"	"	"		"	E7	"				
	$V_{IO}(y)$	8	---	---	+10 V	"	0 V	"	---	"	"	"		"	E8	"	$V_{IO}(y) = \frac{(E8-E9)}{2}$	-50.0	50.0	"
	$V_{IO}(y)$	9	---	---	-10 V	"	"	"	---	"	"	"		"	E9	"				
	$I_{IB}(z)$	10	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	I1	µA	$I_{IB}(z) = I1$	---	6.0	µA
	$+I_{IB}(x)$	11	---	---	---	"	0 V	"	---	"	"	"	K5	8	I2	"		-2.5	2.5	"
	$+I_{IB}(y)$	12	---	---	0 V	"	---	"	---	"	"	"	K6	"	I3	"		"	"	"
	$-I_{IB}(x)$	13	---	---	"	---	0 V	"	---	"	"	"	K2	"	I4	"		"	"	"
	$-I_{IB}(y)$	14	---	---	"	0 V	"	---	---	"	"	"	K4	"	I5	"		"	"	"
	$I_{IO}(x)$	15	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO}(x) = I2\text{-}I4$	-200.0	200.0	nA
	$I_{IO}(y)$	16	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO}(y) = I3\text{-}I5$	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
1 T_A = +25°C	$I_{OS}(+)$	17	---	---	+10 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		11	16	mA	$I_{OS}(+) = 16$	30.0		mA
	$I_{OS}(-)$	18	---	---	"	"	-10 V	"	---	"	"	"		"	17	"	$I_{OS}(-) = 17$		30.0	"
	I_{CC}	19	---	---	0 V	"	0 V	"	---	"	"	"		10	18	"	$I_{CC} = 18$		6.0	"
	I_{EE}	20	---	---	"	"	"	"	---	"	"	"		9	19	"	$I_{EE} = 19$	-6.0		"
	CMRR(x)	21	---	---	-6 V	-6 V	+10 V	"	---	"	"	"		11	E10	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	90.0		dB
	CMRR(x)	22	---	---	0 V	0 V	"	"	---	"	"	"		"	E11	"				
	CMRR(x)	23	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E12	"				
	CMRR(y)	24	---	---	"	0 V	-6 V	-6 V	---	"	"	"		"	E13	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	25	---	---	"	"	0 V	0 V	---	"	"	"		"	E14	"				
	CMRR(y)	26	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E15	"				
	ERR1	27	---	---	-10 V	"	-5 V	0 V	---	"	"	"		"	E16	V		4.9	5.1	V
	ERR2	28	---	---	+10 V	"	"	"	---	"	"	"		"	E17	"		-5.1	-4.9	"
	ERR3	29	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E18	"		"	"	"
	ERR4	30	---	---	+10 V	"	"	"	---	"	"	"		"	E19	"		4.9	5.1	"
	ERR5	31	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E20	"		"	"	"
	ERR6	32	---	---	"	"	+10 V	"	---	"	"	"		"	E21	"		-5.1	-4.9	"
	ERR7	33	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E22	"		"	"	"
	ERR8	34	---	---	"	"	+10 V	"	---	"	"	"		"	E23	"		4.9	5.1	"
	V_{OP}	35	---	---	+12 V	"	"	"	---	"	"	"		"	E24	"	$V_{OP} = E24$	+10.0		"
	V_{OP}	36	---	---	"	"	-10 V	"	---	"	"	"		"	E25	"	$V_{OP} = E25$		-10.0	"
	PSRR1	37	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E26	"	$PSRR1 = \frac{(E26-E27)}{1.5}$	-50.0	50.0	$\frac{mV}{V}$
	PSRR1	38	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E27	"				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
1 T_A = +25°C	PSRR2	39	---	---	+5 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		11	E28	V	$PSRR2 = \frac{(E28-E29)}{3.0}$	40.0	40.0	$\frac{mV}{V}$
	PSRR2	40	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E29	"				
	FT(x)	41	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E30	mV	$FT(x) = \left(\frac{E30+E31}{2}\right) - E5$	-50.0	50.0	mV
	FT(x)	42	---	---	+10 V	"	"	"	---	"	"	"		"	E31	"				
	FT(y)	43	---	---	0 V	"	-10 V	"	---	"	"	"		"	E32	"	$FT(y) = \left(\frac{E32+E33}{2}\right) - E5$	-40.0	40.0	"
	FT(y)	44	---	---	"	"	+10 V	"	---	"	"	"		"	E33	"				
	+NLx(A)	45	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E21-((E2-E4)/4+(((E2+E4)/2)+E33)/2	0.6	0.6	%FS
	+NLx(B)	46	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E23+((E2-E4)/4+(((E2+E4)/2)+E33)/2	"	"	"
	-NLx(A)	47	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E20-((E1-E3)/4+(((E1+E3)/2+E32)/2	"	"	"
	-NLx(B)	48	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E22+((E1-E3)/4+(((E1+E3)/2+E32)/2	"	"	"
	+NLy(A)	49	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E17-((E3-E4)/4+(((E4+E3)/2+E31)/2	0.4	0.4	"
	+NLy(B)	50	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E19+((E3-E4)/4+(((E4+E3)/2+E31)/2	"	"	"
	-NLy(A)	51	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E18-((E2-E1)/4+(((E2+E1)/2+E30)/2	"	"	"
	-NLy(B)	52	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E16+((E2-E1)/4+(((E2+E1)/2+E30)/2	"	"	"
	+NLx	53	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E4+E2)/2)-E33)/2	0.6	0.6	"
	-NLx	54	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E3+E1)/2)-E32)/2	"	"	"
	+NLy	55	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E4+E3)/2)-E31)/2	0.4	0.4	"
	-NLy	56	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E1+E2)/2)-E30)/2	"	"	"
2 T_A = +125°C	MAx-y-	57	---	---	-10 V	0 V	-10 V	0 V	---	"	-15 V	+15 V		11	E34	V	$MAx\text{-}y\text{-} = \frac{(E34-10)}{10} \times 100$	4.0	4.0	"
	MAx-y+	58	---	---	"	"	+10 V	"	---	"	"	"		"	E35	"	$MAx\text{-}y+ = \frac{(E35+10)}{10} \times 100$	"	"	"
	MAx+y-	59	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E36	"	$MAx+y\text{-} = \frac{(E36+10)}{10} \times 100$	"	"	"
	MAx+y+	60	---	---	"	"	+10 V	"	---	"	"	"		"	E37	"	$MAx+y+ = \frac{(E37-10)}{10} \times 100$	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
2 T_A = +125°C	$\frac{MAxy}{\Delta T}$	61	---	---	+10 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		---	---	---	$\frac{MAxy}{\Delta T} = \frac{(MAxy\ (125°C) - MAxy\ (25°C)}{100}$	-0.04	0.04	%FS
	V_{OIO}	62	---	---	0 V	"	0 V	"	---	"	"	"		11	E38	mV	$V_{OIO} = E38$	-100.0	100.0	mV
	$\frac{V_{OIO}}{\Delta T}$	63	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{OIO}}{\Delta T} = \frac{(E38-E5)}{100}$	-1.0	1.0	$\frac{mV}{°C}$
	$V_{IO(x)}$	64	---	---	"	"	-10 V	"	---	"	"	"		11	E39	mV	$V_{IO(x)} = \frac{(E39-E40)}{2}$	-90.0	90.0	mV
	$V_{IO(x)}$	65	---	---	"	"	+10 V	"	---	"	"	"		"	E40	"				
	$\frac{V_{IO(x)}}{\Delta T}$	66	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(x)}}{\Delta T} = \frac{(V_{IO(x)}\ @125°C - V_{IO(x)}\ @25°C)}{100}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$V_{IO(y)}$	67	---	---	-10 V	"	0 V	"	---	"	"	"		11	E41	mV	$V_{IO(y)} = \frac{(E41-E42)}{2}$	-100.0	100.0	mV
	$V_{IO(y)}$	68	---	---	+10 V	"	"	"	---	"	"	"		"	E42	"				
	$\frac{V_{IO(y)}}{\Delta T}$	69	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO(y)}}{\Delta T} = \frac{(V_{IO(y)}\ @125°C - V_{IO(y)}\ @25°C)}{100}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$I_{IB(z)}$	70	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	I10	µA	$I_{IB(z)} = I10$	---	6.0	µA
	$+I_{IB(x)}$	71	---	---	---	"	0 V	"	---	"	"	"	K5	8	I11	"		-2.5	2.5	"
	$+I_{IB(y)}$	72	---	---	0 V	"	---	"	---	"	"	"	K6	"	I12	"		"	"	"
	$-I_{IB(x)}$	73	---	---	"	---	0 V	"	---	"	"	"	K2	"	I13	"		"	"	"
	$-I_{IB(y)}$	74	---	---	"	0 V	"	"	---	"	"	"	K4	"	I14	"		"	"	"
	$I_{IO(x)}$	75	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO(x)} = I11-I13$	-400.0	400.0	nA
	$I_{IO(y)}$	76	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO(y)} = I12-I14$	"	"	"
	$I_{OS(+)}$	77	---	---	+10 V	"	+10 V	"	---	"	"	"		11	I15	mA	$I_{OS(+)} = I15$	-30.0		mA
	$I_{OS(-)}$	78	---	---	"	"	-10 V	"	---	"	"	"		"	I16	"	$I_{OS(-)} = I16$		30.0	"
	I_{CC}	79	---	---	0 V	"	0 V	"	---	"	"	"		10	I17	"	$I_{CC} = I17$		6.0	"
	I_{EE}	80	---	---	"	"	"	"	---	"	"	"		9	I18	"	$I_{EE} = I18$	-6.0		"

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
2 T_A = +125°C	CMRR(x)	81	---	---	-6 V	-6 V	+10 V	0 V	---	GND	-15 V	+15 V		11	E43	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	90.0		dB
	CMRR(x)	82	---	---	0 V	0 V	"	"	---	"	"	"		"	E44	"				
	CMRR(x)	83	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E45	"				
	CMRR(y)	84	---	---	"	0 V	-6 V	-6 V	---	"	"	"		"	E46	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	85	---	---	"	"	0 V	0 V	---	"	"	"		"	E47	"				
	CMRR(y)	86	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E48	"				
	ERR1	87	---	---	-10 V	"	-5 V	0 V	---	"	"	"		"	E49	V		4.6	5.4	V
	ERR2	88	---	---	+10 V	"	"	"	---	"	"	"		"	E50	"		-5.4	-4.6	"
	ERR3	89	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E51	"		"	"	"
	ERR4	90	---	---	+10 V	"	"	"	---	"	"	"		"	E52	"		4.6	5.4	"
	ERR5	91	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E53	"		"	"	"
	ERR6	92	---	---	"	"	+10 V	"	---	"	"	"		"	E54	"		-5.4	-4.6	"
	ERR7	93	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E55	"		"	"	"
	ERR8	94	---	---	"	"	+10 V	"	---	"	"	"		"	E56	"		4.6	5.4	"
	V_{OP}	95	---	---	+12 V	"	"	"	---	"	"	"		"	E57	"	V_{OP} = E57	+10.0		"
	V_{OP}	96	---	---	"	"	-10 V	"	---	"	"	"		"	E58	"	V_{OP} = E58		-10.0	"
	PSRR1	97	---	---	+5 V	"	+10 V	"	---	"	-13.5 V	+13.5 V		"	E59	"	$PSRR1 = \frac{(E59-E60)}{1.5}$	-50.0	50.0	mV/V
	PSRR1	98	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E60	"				
	PSRR2	99	---	---	"	"	"	"	---	"	"	"		"	E61	"	$PSRR2 = \frac{(E61-E62)}{3.0}$	-40.0	40.0	"
	PSRR2	100	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E62	"				
	FT(x)	101	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E63	mV	$FT(x) = \left(\frac{E63+E64}{2}\right) - E38$	-50.0	50.0	mV
	FT(x)	102	---	---	+10 V	"	"	"	---	"	"	"		"	E64	"				
	FT(y)	103	---	---	0 V	"	-10 V	"	---	"	"	"		"	E65	"	$FT(y) = \left(\frac{E65+E66}{2}\right) - E38$	40.0	40.0	"
	FT(y)	104	---	---	"	"	+10 V	"	---	"	"	"		"	E66	"				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equations	Limits Min	Max	Unit
2 T_A = +125°C	+NLx(A)	105	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E54-((E35-E37)/4+(((E35+E37)/2)+E66)/2	-0.6	0.6	%FS
	+NLx(B)	106	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E56+((E35-E37)/4+(((E35+E37)/2)+E66)/2	"	"	"
	-NLx(A)	107	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E53-((E34-E36)/4+(((E34+E36)/2)+E65)/2	"	"	"
	-NLx(B)	108	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E55+((E34-E36)/4+(((E34+E36)/2)+E65)/2	"	"	"
	+NLy(A)	109	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E50-((E36-E37)/4+(((E36+E37)/2)+E64)/2	-0.4	0.4	"
	+NLy(B)	110	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E52+((E36-E37)/4+(((E36+E37)/2)+E64)/2	"	"	"
	-NLy(A)	111	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E51-((E35-E34)/4+(((E35+E34)/2)+E63)/2	"	"	"
	-NLy(B)	112	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E49+((E35-E34)/4+(((E35+E34)/2)+E63)/2	"	"	"
	+NLx	113	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E37+E35)/2)-E66)/2	0.6	0.6	"
	-NLx	114	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E36+E34)/2-E65)/2	"	"	"
	+NLy	115	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E37+E36)/2)-E64)/2	-0.4	0.4	"
	-NLy	116	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy = (((E34+E35)/2)-E63)/2	"	"	"
3 T_A = -55°C	MAx-y-	117	---	---	-10 V	0 V	-10 V	0 V	---	GND	-15 V	+15 V		11	E67	V	$MAx\text{-}y\text{-} = \frac{(E67-10)}{10} \times 100$	4.0	4.0	"
	MAx-y+	118	---	---	"	"	+10 V	"	---	"	"	"		"	E68	"	$MAx\text{-}y+ = \frac{(E68+10)}{10} \times 100$	"	"	"
	MAx+y-	119	---	---	+10 V	"	-10 V	"	---	"	"	"		"	E69	"	$MAx+y\text{-} = \frac{(E69+E10)}{10} \times 100$	"	"	"
	MAx+y+	120	---	---	"	"	+10 V	"	---	"	"	"		"	E70	"	$MAx+y+ = \frac{(E70-10)}{10} \times 100$	"	"	"
	$\frac{MAxy}{\Delta T}$	121	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{MAxy}{\Delta T} = \left[\frac{(MAxy\ (-55°C)-MAxy\ (25°C)}{80}\right]$	0.04	0.04	$\frac{\%FS}{\Delta T}$
	V_{OIO}	122	---	---	0 V	"	0 V	"	---	"	"	"		11	E71	mV	V_{OIO} = E71	-100.0	100.0	mV
	$\frac{V_{OIO}}{\Delta T}$	123	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{OIO}}{\Delta T} = \frac{(E71-E5)}{80}$	-1.0	1.0	$\frac{mV}{°C}$
	$V_{IO}(x)$	124	---	---	"	"	-10 V	"	---	"	"	"		11	E72	mV	$V_{IO}(x) = \frac{(E72-E73)}{2}$	-90.0	90.0	mV
	$V_{IO}(x)$	125	---	---	"	"	+10 V	"	---	"	"	"		"	E73	"				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
3 T_A = -55°C	$\frac{V_{IO}(x)}{\Delta T}$	126	---	---	0 V	0 V	+10 V	0 V	---	GND	-15 V	+15 V		---	---	---	$\frac{V_{IO}(x)}{\Delta T} = \frac{(V_{IO}(x)\ @\ -55°C - V_{IO}(x)\ @\ 25°C)}{80}$	-100.0	100.0	$\frac{\mu V}{°C}$
	$V_{IO}(y)$	127	---	---	-10 V	"	0 V	"	---	"	"	"		11	E74	mV	$V_{IO}(y) = \frac{(E74 - E75)}{2}$	-100.0	100.0	mV
	$V_{IO}(y)$	129	---	---	+10 V	"	"	"	---	"	"	"		"	E75	"				
	$\frac{V_{IO}(y)}{\Delta T}$	129	---	---	"	"	"	"	---	"	"	"		---	---	---	$\frac{V_{IO}(y)}{\Delta T} = \frac{(V_{IO}(y)\ @\ -55°C - V_{IO}(y)\ @\ 25°C)}{80}$	-800.0	800.0	$\frac{\mu V}{°C}$
	$I_{IB}(z)$	130	---	---	+5 V	"	---	"	---	"	"	"	K8, K9	7	I19	µA	$I_{IB}(z) = I19$	---	6.0	µA
	$+I_{IB}(x)$	131	---	---	---	"	0 V	"	---	"	"	"	K5	8	I20	"		6.0	"	"
	$+I_{IB}(y)$	132	---	---	0 V	"	---	"	---	"	"	"	K6	"	I21	"		"	"	"
	$-I_{IB}(x)$	133	---	---	"	---	0 V	"	---	"	"	"	K2	"	I22	"		"	"	"
	$-I_{IB}(y)$	134	---	---	"	0 V	"	---	---	"	"	"	K4	"	I23	"		"	"	"
	$I_{IO}(x)$	135	---	---	"	"	"	0 V	---	"	"	"		---	---	---	$I_{IO}(x) = I20 - I22$	-400.0	400.0	nA
	$I_{IO}(y)$	136	---	---	"	"	"	"	---	"	"	"		---	---	---	$I_{IO}(y) = I21 - I23$	"	"	"
	$I_{OS}(+)$	137	---	---	+10 V	"	+10 V	"	---	"	"	"		11	I24	mA	$I_{OS}(+) = I24$	-30.0		mA
	$I_{OS}(-)$	138	---	---	"	"	-10 V	"	---	"	"	"		"	I25	"	$I_{OS}(-) = I25$		30.0	"
	I_{CC}	139	---	---	0 V	"	0 V	"	---	"	"	"		10	I26	"	$I_{CC} = I26$		6.0	"
	I_{EE}	140	---	---	"	"	"	"	---	"	"	"		9	I27	"	$I_{EE} = I27$	-6.5		"
	CMRR(x)	141	---	---	-6 V	-6 V	+10 V	"	---	"	"	"		11	E76	mV	$CMRR(x) = 20 \log\left(\frac{10}{\Delta E_x}\right)$	80		dB
	CMRR(x)	142	---	---	0 V	0 V	"	"	---	"	"	"		"	E77	"				
	CMRR(x)	143	---	---	+10 V	+10 V	"	"	---	"	"	"		"	E78	"				
	CMRR(y)	144	---	---	"	0 V	-6 V	-6 V	---	"	"	"		"	E79	"	$CMRR(y) = 20 \log\left(\frac{10}{\Delta E_y}\right)$	"		"
	CMRR(y)	145	---	---	"	"	0 V	0 V	---	"	"	"		"	E80	"				
	CMRR(y)	146	---	---	"	"	+10 V	+10 V	---	"	"	"		"	E81	"				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
3 T_A = -55°C	ERR1	147	---	---	-10 V	0 V	-5 V	0 V	---	GND	-15 V	+15 V		11	E82	V		4.6	5.4	V
	ERR2	148	---	---	+10 V	"	"	"	---	"	"	"		"	E83	"		5.4	4.6	"
	ERR3	149	---	---	-10 V	"	+5 V	"	---	"	"	"		"	E84	"		"	"	"
	ERR4	150	---	---	+10 V	"	"	"	---	"	"	"		"	E85	"		4.6	5.4	"
	ERR5	151	---	---	-5 V	"	-10 V	"	---	"	"	"		"	E86	"		"	"	"
	ERR6	152	---	---	"	"	+10 V	"	---	"	"	"		"	E87	"		5.4	4.6	"
	ERR7	153	---	---	+5 V	"	-10 V	"	---	"	"	"		"	E88	"		"	"	"
	ERR8	154	---	---	"	"	+10 V	"	---	"	"	"		"	E89	"		4.6	5.4	"
	V_{OP}	155	---	---	+15 V	"	"	"	---	"	"	"		"	E90	"	V_{OP} = E90	+10.0		"
	V_{OP}	156	---	---	"	"	-10 V	"	---	"	"	"		"	E91	"	V_{OP} = E91		-10.0	"
	PSRR1	157	---	---	"	"	"	"	---	"	-13.5 V	+13.5 V		"	E92	"	$PSRR1 = \frac{(E92-E93)}{1.5}$	-50.0	50.0	$\frac{mV}{V}$
	PSRR1	158	---	---	"	"	"	"	---	"	-15 V	+15 V		"	E93	"				
	PSRR2	159	---	---	"	"	"	"	---	"	"	"		"	E94	"	$PSRR2 = \frac{(E94-E95)}{3.0}$	-40.0	40.0	"
	PSRR2	160	---	---	"	"	"	"	---	"	-18 V	+18 V		"	E95	"				
	FT(x)	161	---	---	-10 V	"	0 V	"	---	"	-15 V	+15 V		"	E96	mV	$FT(x) = \left(\frac{E96+E97}{2}\right) - E71$	-50.0	50.0	mV
	FT(x)	162	---	---	+10 V	"	"	"	---	"	"	"		"	E97	"				
	FT(y)	163	---	---	0 V	"	-10 V	"	---	"	"	"		"	E98	"	$FT(y) = \left(\frac{E98+E99}{2}\right) - E71$	-40.0	40.0	"
	FT(y)	164	---	---	"	"	+10 V	"	---	"	"	"		"	E99	"				
	+NLx(A)	165	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(A) = E87-(((E68-E70)/4+(((E68+E70)/2)+E99)/2	-0.6	0.6	%FS
	+NLx(B)	166	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx(B) = E89+(((E68-E70)/4+(((E68+E70)/2)+E99)/2	"	"	"
	-NLx(A)	167	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(A) = E86-(((E67-E69)/4+(((E67+E69)/2)+E98)/2	"	"	"
	-NLx(B)	168	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx(B) = E88+(((E67-E69)/4+(((E67+E69)/2)+E98)/2	"	"	"
	+NLy(A)	169	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(A) = E83-(((E69-E70)/4+(((E69+E70)/2)+E97)/2	0.4	0.4	"
	+NLy(B)	170	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy(B) = E85+(((E69-E70)/4+(((E69+E70)/2)+E97)/2	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
3 T_A = -55°C	-NLy(A)	171	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(A) = E84-(((E68-E67)/4+(((E68+E67)/2)+E96)/2	0.4	0.4	% FS
	-NLy(B)	172	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLy(B) = E82+(((E68-E67)/4+(((E68+E67)/2)+E96)/2	"	"	"
	+NLx	173	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLx = (((E70+E68)/2)-E99)/2	0.6	0.6	"
	-NLx	174	---	---	---	---	---	---	---	---	---	---		---	---	---	-NLx = (((E69+E67)/2)-E98)/2	"	"	"
	+NLy	175	---	---	---	---	---	---	---	---	---	---		---	---	---	+NLy = (((E70+E69)/2)-E97)/2	0.4	0.4	"
	-NLy	176	---	---	---	---	---	---	---	---	---	---					-NLy = (((E67+E68)/2)-E96)/2	"	"	"
4 T_A = +25°C	SR(+)	177	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	20.0		$\frac{V}{\mu s}$
	SR(+)	178	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	µs				
	SR(-)	179	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	"		"
	SR(-)	180	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	µs				
	AE(x)	181	2 Vp-p	---	---	"	"	"	---	"	"	"	K1, K12, K13, K14	"	f1	kHz	AE(x) = f1 @(-1%) Eo	70.0		kHz
	AE(y)	182	"	---	+10 V	"	---	"	---	"	"	"	"	"	f2	"	AE(y) = f2 @(-1%) Eo	"		"
5 T_A = +125°C	SR(+)	183	---	20 Vp-p f = 10kHz	---	"	+10 V	"	---	"	"	"	K1, K12, K15	"	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	20.0		$\frac{V}{\mu s}$
	SR(+)	184	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	µs				
	SR(-)	185	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	"		"
	SR(-)	186	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	µs				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers										Energized relays	Measured pin			Equations	Limits		Unit
			1	2	3	4	5	6	7	8	9	10		No.	Value	Unit		Min	Max	
6 T_A = -55°C	SR(+)	187	---	20 Vp-p f = 10kHz	---	0 V	+10 V	0 V	---	GND	-15 V	+15 V	K1, K12, K15	11	ΔVo (+)	V	See figure 5, waveform 1 SR(+) = ΔVo(+)/Δt(+)	20.0		$\frac{V}{\mu s}$
	Δt(+)	188	---	"	---	"	"	"	---	"	"	"	"	"	Δt(+)	μs				
	SR(-)	189	---	"	---	"	"	"	---	"	"	"	"	"	ΔVo (-)	V	See figure 5, waveform 2 SR(-) = ΔVo(-)/Δt(-)	"		"
	Δt(-)	190	---	"	---	"	"	"	---	"	"	"	"	"	Δt(-)	μs				
7 T_A = +25°C	N1(BB)	191	---	---	--	---	---	---	---	---	---	---		---	---	---	See figure 6 BW = 10 Hz to 100 kHz		3.0	mVrms
9 T_A = +25°C	ts(+)	192	---	20 Vp-p f = 10kHz	---	0 V	+10 V	0 V	---	GND	-15 V	+15 V	K1, K12, K15	11	ts(+)	μs	See figure 5, waveform 1		2.0	μs
	ts(+)	193	---	"	---	"	"	"	---	"	"	"	"	"	ts(-)	"	See figure 5, waveform 2		"	"

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Class S devices - Test condition A, using the circuit shown on figure 3.

(2) Class B devices - Test condition A using the circuit shown on figure 3.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). This screen test shall apply to class S devices only using the circuit shown on figure 3.

c. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

e. Internal visual inspection methods 2010 and 2017 (hybrid and multichip criteria) of MIL-STD-883. For multichip devices, internal visual inspection shall be performed for each chip within the package.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3.1 Qualification extension. For qualification inspection, if a manufacturer qualifies to device type 01, which is designed and manufactured identically (same die, same process, same screening) in all respects (except electrical testing) to device type 02, then qualification may be extended to device type 02 when authorized by the qualifying activity. Additionally, part I qualification may be extended to device type 02 only after acceptance by the qualifying activity of subgroup C1 testing performed on the S device type and submission of data in accordance with MIL-M-38510, appendix D.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 8, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 3 herein.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady-state life test for class B (method 1005 of MIL-STD-883. Test condition D using the circuit shown on figure 3, test condition C using the circuit shown on figure 4, or test condition F using the circuit shown on figure 6.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltage values are referenced to the external zero reference level of the supply voltage. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled within 10°C of their power stable condition at room temperature prior to removal of the bias.

4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be applied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2c).

TABLE IV. Group C end-point electrical parameters.
(T_A = 25°C, V_S = ±15 V dc, V_{OS} = GND.)

Table III test no.	Test	Unit	01 delta		02 delta		03 delta		04 delta	
			Min	Max	Min	Max	Min	Max	Min	Max
1	MAx-y-	%FS	-0.1	0.1	-0.1	0.1	-0.2	0.2	-0.2	0.2
2	MAx-y+	%FS	-0.1	0.1	-0.1	0.1	-0.2	0.2	-0.2	0.2
3	MAx+y-	%FS	-0.1	0.1	-0.1	0.1	-0.2	0.2	-0.2	0.2
4	MAx+y+	%FS	-0.1	0.1	-0.1	0.1	-0.2	0.2	-0.2	0.2
11,12,13	$+I_{IB}$	µA	-0.2	0.2	-0.2	0.2	---	---	---	---
14,15,16	$-I_{IB}$	µA	-0.2	0.2	-0.2	0.2	---	---	---	---
11,12	$+I_{IB}$	µA	---	---	---	---	-0.2	0.2	-0.2	0.2
13,14	$-I_{IB}$	µA	---	---	---	---	-0.2	0.2	-0.2	0.2

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for original equipment and logistic support of existing design application equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	534T
02	534S
03	532S
04	4213

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 80, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Civil agencies coordinating activity:
NASA - NA

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/1393	MICROCIRCUITS, LINEAR, INTERNALLY TRIMMED ANALOG MULTIPLIER, MONOLITHIC AND HYBRID, SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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